File :....71_constructie_advies\2992-6.2 Portaal.xfr2

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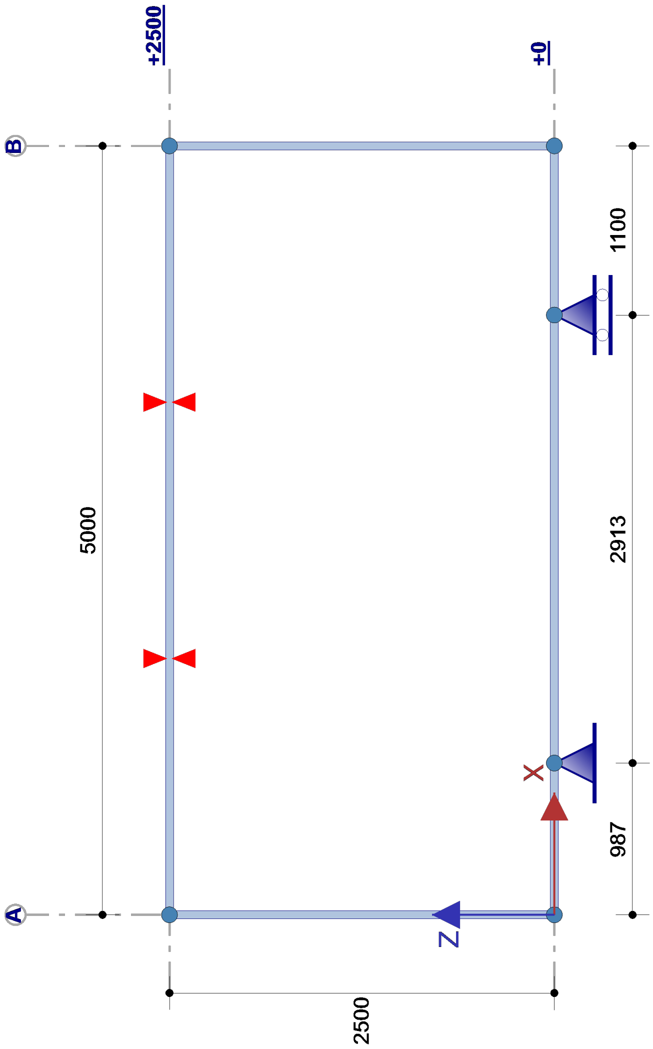
Applied standards : NEN-EN 1993-1-1+C2+A1/NB:2016 nl
Consequence class : CC1
Gravity acceleration g : 9,81 m/s2

1 Input Data



1.1 NODES

Node Number	Coordinates		Restraints		
	X [mm]	Z [mm]	Tx	Tz	Ry
1	0	0			
2	987	0	A	A	
3	3900	0		A	
4	5000	0			
5	0	2500			
6	5000	2500			

1.2 BEAMS

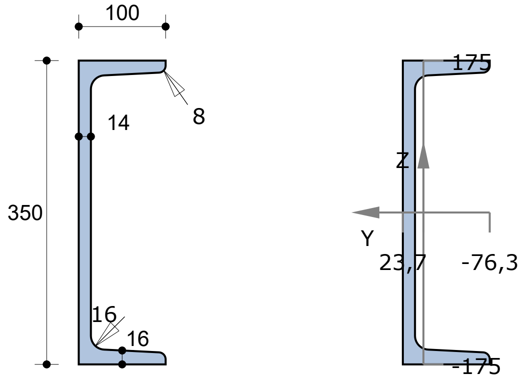
Beam Number	Node		Beam type	Profile	Length [mm]
	from	to			
1	1	4		UNP350	5000
2	1	5		HEB160	2500
3	4	6		HEB160	2500

Beam Number	Node from	Node to	Beam type	Profile	Length [mm]
4	5	6	⊢—⊣	HEB300	5000

1.3 PROFILES

Profile Number	Name	Weight [kg/m]	E [N/mm²]	A [mm²]	Iy [mm4]	Wy;el_1 [mm³]	Wy;el_2 [mm³]
1	UNP350	60,2	210000	7,666E3	1,2694E8	7,2538E5	7,2538E5
2	HEB160	42,6	210000	5,428E3	2,4929E7	3,1161E5	3,1161E5
4	HEB300	117,0	210000	1,491E4	2,5169E8	1,6779E6	1,6779E6

UNP350



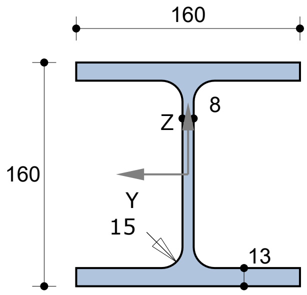
Material data

Steel grade	S 235 Hot rolled	
Elasticity Modulus	E =	210000 N/mm2

Cross section data

Maximum coordinate	y_{max} =	23,7 mm	z_{max} =	175,0 mm
Minimum coordinate	y_{min} =	-76,3 mm	z_{min} =	-175,0 mm
Centroid	z_s =	0,0 mm	y_s =	0,0 mm
Area / Weight	A =	7665,7 mm^2	G =	60,2 kg/m
First moment of area	S_y =	444882 mm^3	S_z =	84137 mm^3
Moment of inertia	I_y =	126942139 mm^4	I_z =	5603739 mm^4
Radius of gyration	i_y =	128,7 mm	i_z =	27,0 mm
Elastic section modulus	$W_{y;el}$ =	725384 mm^3	$W_{z;el}$ =	73447 mm^3
Product moment of area	C_{yz} =	0 mm^3	angle =	0,00 °
Moment of inertia	I_{max} =	126942139 mm^4	I_{min} =	5603739 mm^4
Radius of gyration	i_{max} =	128,7 mm	i_{min} =	27,0 mm
Plastic neutral axis(PNA)	z_h =	0,0 mm	y_h =	12,8 mm
Plastic section modulus	$W_{y;pl}$ =	889763 mm^3	$W_{z;pl}$ =	139730 mm^3

HEB160



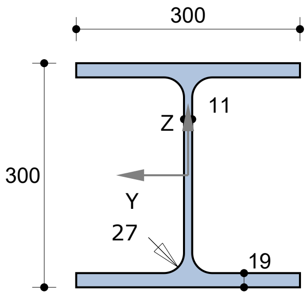
Material data

Steel grade	S 235 Hot rolled		
Elasticity Modulus	E	=	210000 N/mm2

Cross section data

Maximum coordinate	y_{max}	=	80,0 mm	z_{max}	=	80,0 mm
Minimum coordinate	y_{min}	=	-80,0 mm	z_{min}	=	-80,0 mm
Centroid	z_s	=	0,0 mm	y_s	=	0,0 mm
Area / Weight	A	=	5427,5 mm ^{2}	G	=	42,6 kg/m
First moment of area	S_y	=	177057 mm ^{3}	S_z	=	84993 mm ^{3}
Moment of inertia	I_y	=	24929151 mm ^{4}	I_z	=	8892613 mm ^{4}
Radius of gyration	i_y	=	67,8 mm	i_z	=	40,5 mm
Elastic section modulus	$W_{y;el}$	=	311614 mm ^{3}	$W_{z;el}$	=	111158 mm ^{3}
Product moment of area	C_{yz}	=	0 mm ^{3}	angle	=	0,00 °
Moment of inertia	I_{max}	=	24929151 mm ^{4}	I_{min}	=	8892613 mm ^{4}
Radius of gyration	i_{max}	=	67,8 mm	i_{min}	=	40,5 mm
Plastic neutral axis(PNA)	z_h	=	0,0 mm	y_h	=	0,0 mm
Plastic section modulus	$W_{y;pl}$	=	354113 mm ^{3}	$W_{z;pl}$	=	169986 mm ^{3}

HEB300



Material data

Steel grade	S 235 Hot rolled		
Elasticity Modulus	E	=	210000 N/mm2

Cross section data

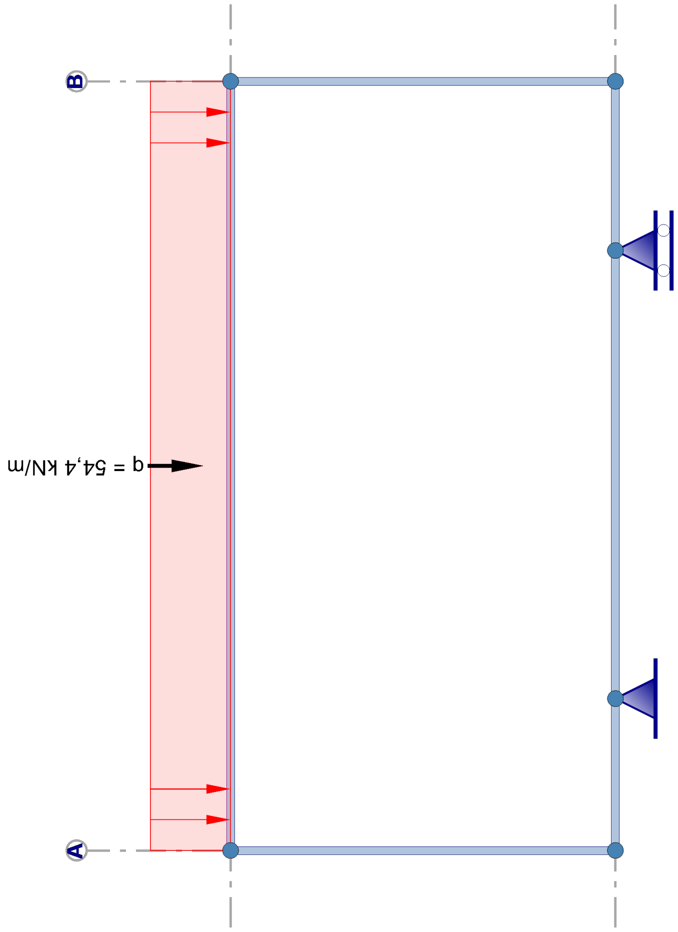
Maximum coordinate	y_{max}	=	150,0 mm	z_{max}	=	150,0 mm
Minimum coordinate	y_{min}	=	-150,0 mm	z_{min}	=	-150,0 mm
Centroid	z_s	=	0,0 mm	y_s	=	0,0 mm
Area / Weight	A	=	14909,9 mm ^{2}	G	=	117,0 kg/m

First moment of area	S_y	=	934466 mm ^{3}	S_z	=	435087 mm ^{3}
Moment of inertia	I_y	=	251688299 mm ^{4}	I_z	=	85628952 mm ^{4}
Radius of gyration	i_y	=	129,9 mm	i_z	=	75,8 mm
Elastic section modulus	$W_{y;el}$	=	1677922 mm ^{3}	$W_{z;el}$	=	570860 mm ^{3}
Product moment of area	C_{yz}	=	0 mm ^{3}	angle	=	0,00 °
Moment of inertia	I_{max}	=	251688299 mm ^{4}	I_{min}	=	85628952 mm ^{4}
Radius of gyration	i_{max}	=	129,9 mm	i_{min}	=	75,8 mm
Plastic neutral axis(PNA)	z_h	=	0,0 mm	y_h	=	0,0 mm
Plastic section modulus	$W_{y;pl}$	=	1868933 mm ^{3}	$W_{z;pl}$	=	870174 mm ^{3}

1.4 LOAD CASES

no.	Description	Type	ψ0	ψ1	ψ2
1	Dead load	Dead load incl. self-weight	1,00	1,00	1,00
2	Live load	A:domestic	0,40	0,50	0,30

1.5 LOAD CASE 1 Dead load Including self-weight



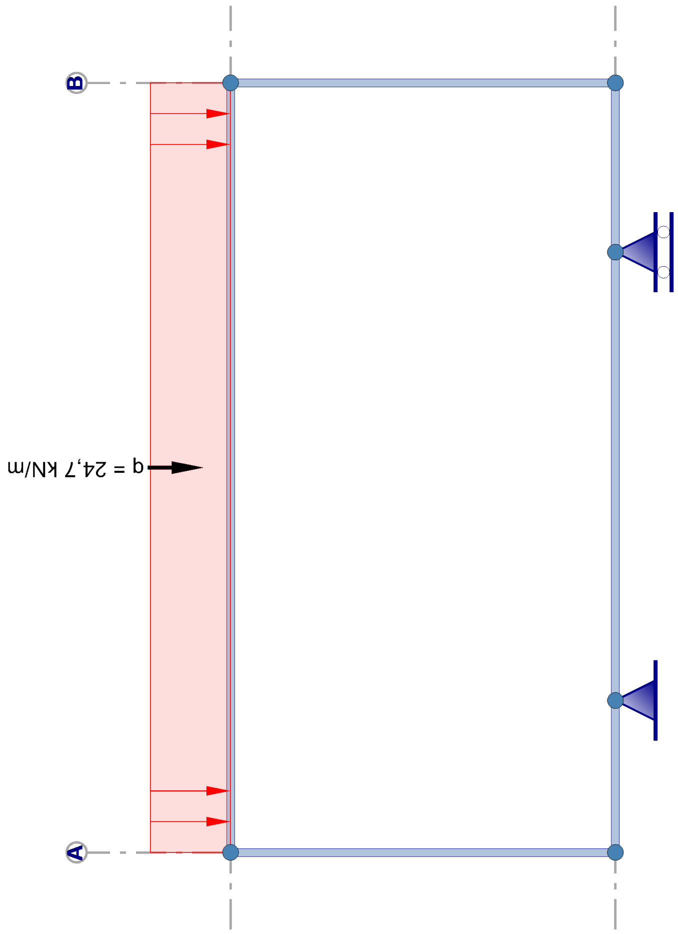
***) Loads due to self-weight are not drawn!**

Total self-weight: : 869 kg.

1.5.1 Beam loads

Beam Number	Loads				Distance from		
	Type	q1	q2	Angle	Node	a [mm]	L [mm]
1	q	-0,590 kN/m	-0,590 kN/m	0,0	1	0	5000
4	q	-1,148 kN/m	-1,148 kN/m	0,0	5	0	5000
4	q	-54,400 kN/m	-54,400 kN/m	0,0	5	0	5000

1.6 LOAD CASE 2 Live load



1.6.1 Beam loads

Beam Number	Loads				Distance from		
	Type	q1	q2	Angle	Node	a [mm]	L [mm]
4	q	-24,700 kN/m	-24,700 kN/m	0,0	5	0	5000

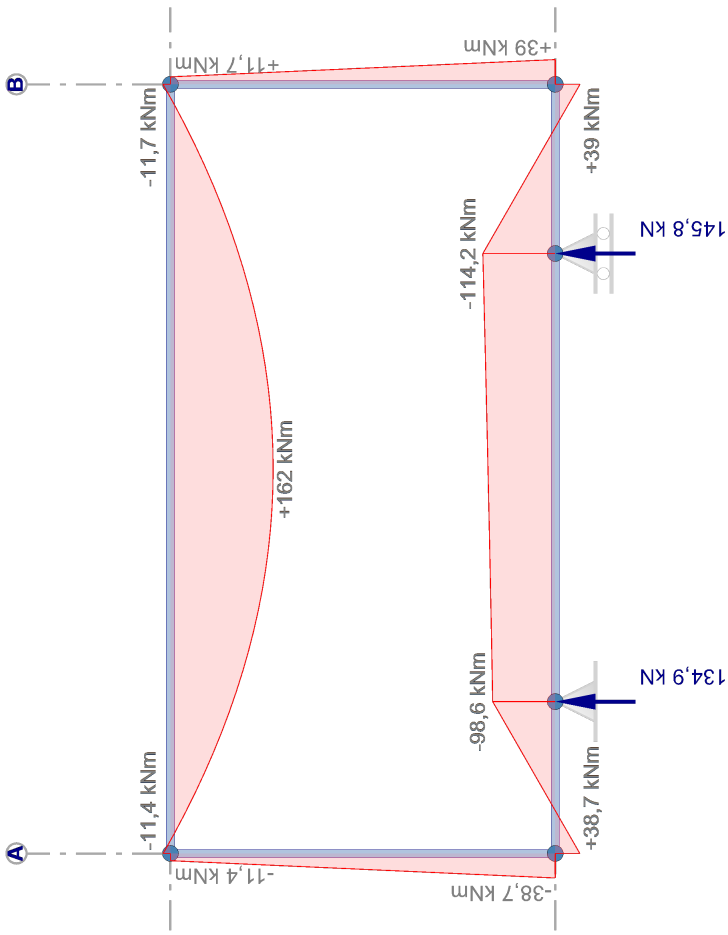
2 Calculation Results

2.1 NODES - Global sway imperfection

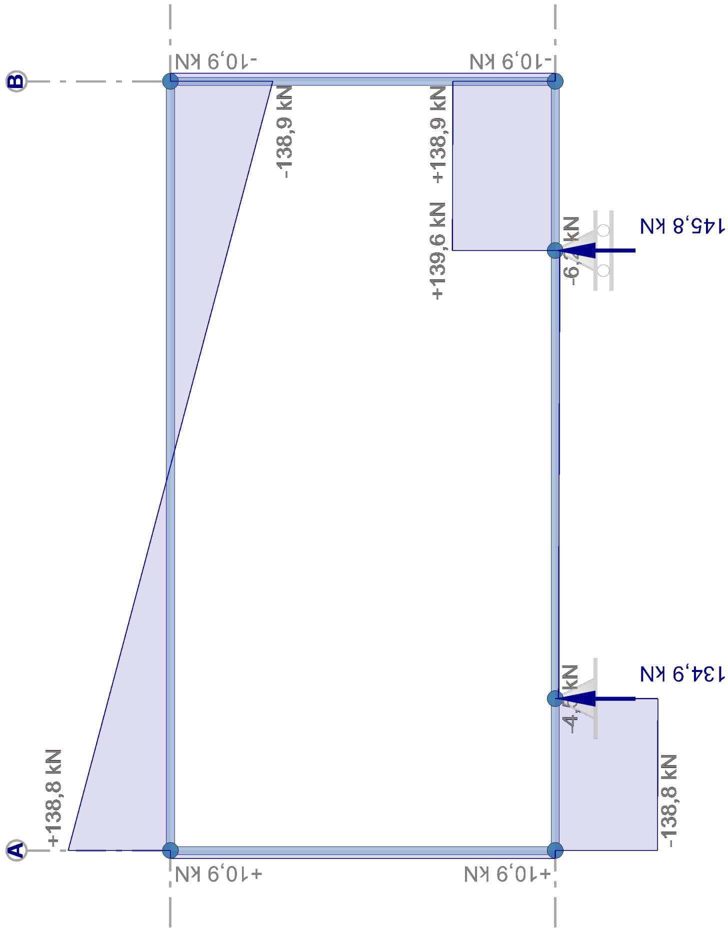
Node Number	1/200 in +X			1/200 in -X	
	X [mm]	Z [mm]		X [mm]	Z [mm]
1	0	0		0	0
2	987	0		987	0
3	3900	0		3900	0
4	5000	0		5000	0
5	13	2500		-13	2500
6	5013	2500		4988	2500

2.2 LOAD CASES

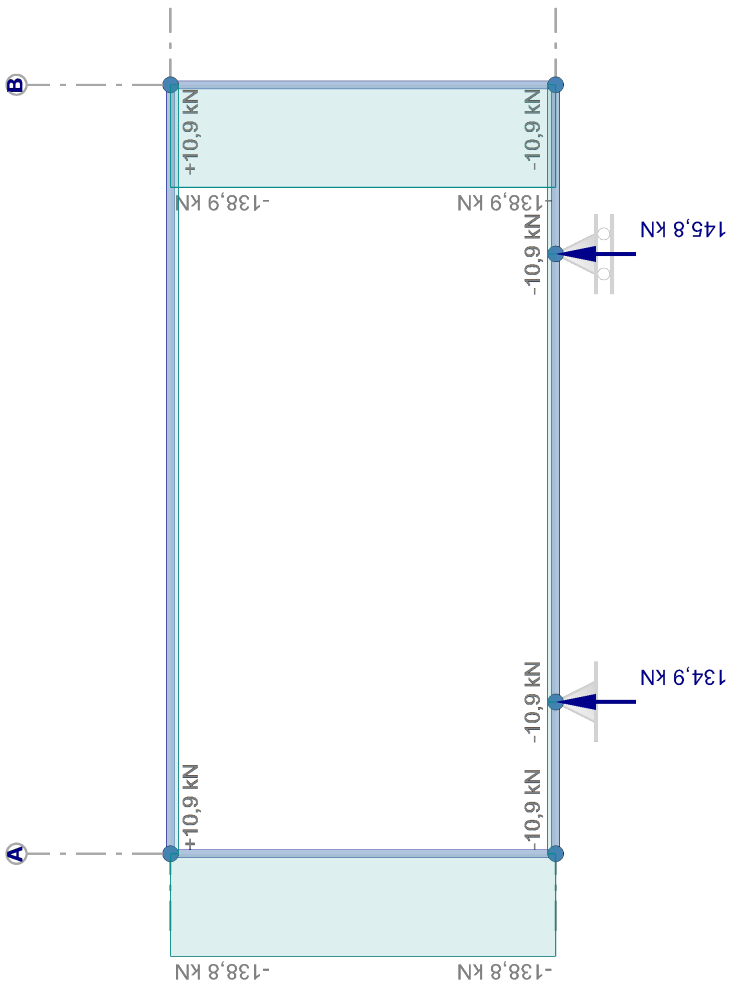
Geometric linear analysis



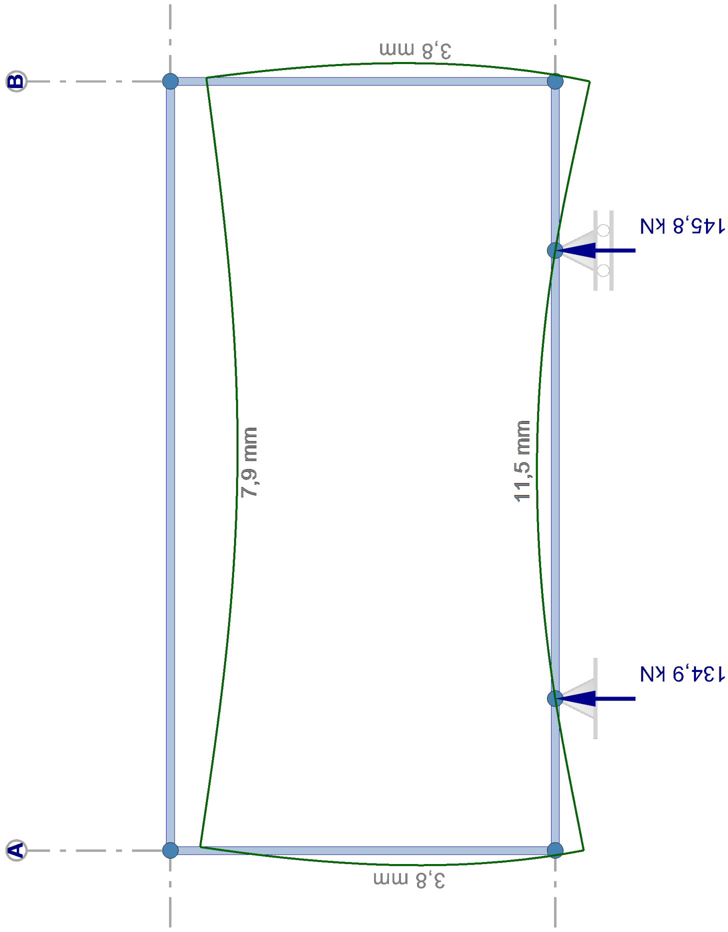
M-line - 1 Dead load


B
A
-10,9 kN
-10,9 kN
-138,9 kN
+138,9 kN
+139,6 kN
145,8 kN
134,9 kN
+138,8 kN
-138,8 kN
+10,9 kN
+10,9 kN

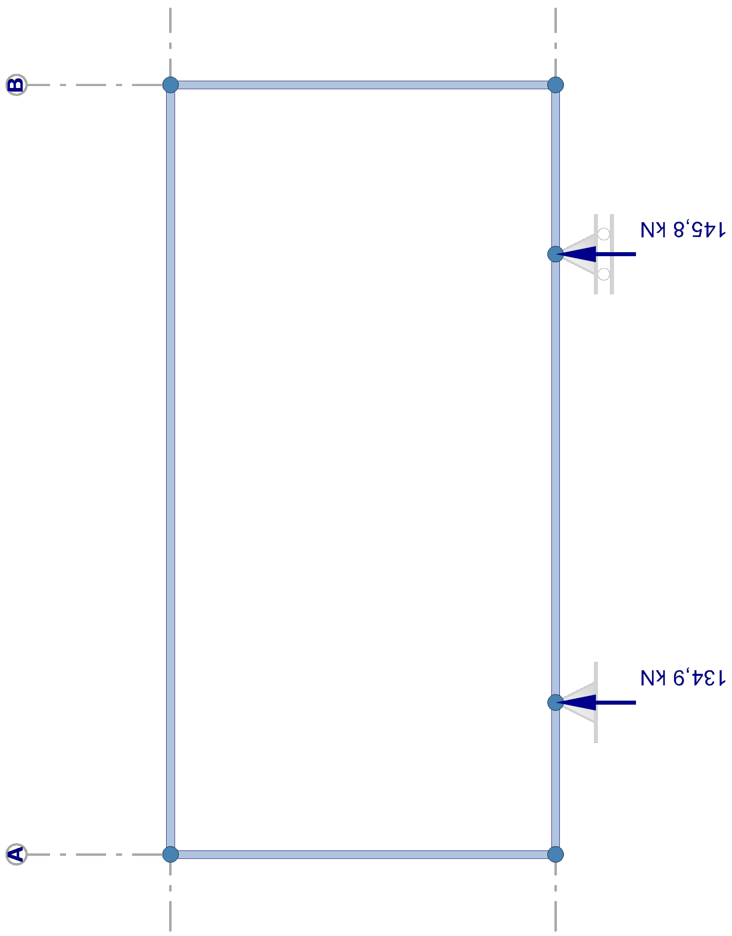
D-line - 1 Dead load



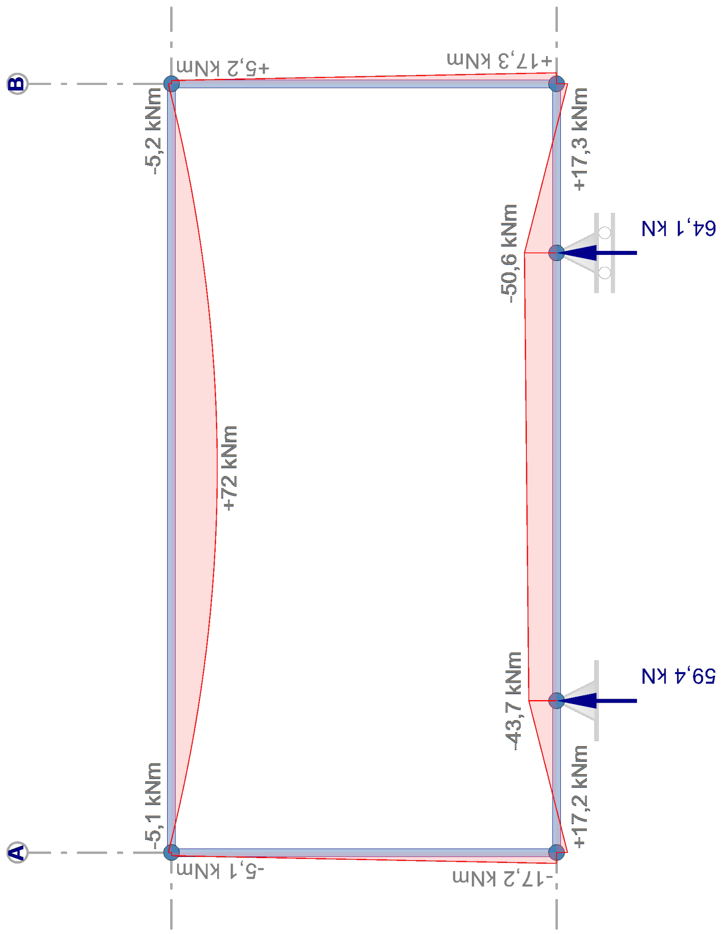
N-line - 1 Dead load


B
A
3,8 mm
7,9 mm
11,5 mm
145,8 kN
134,9 kN
3,8 mm

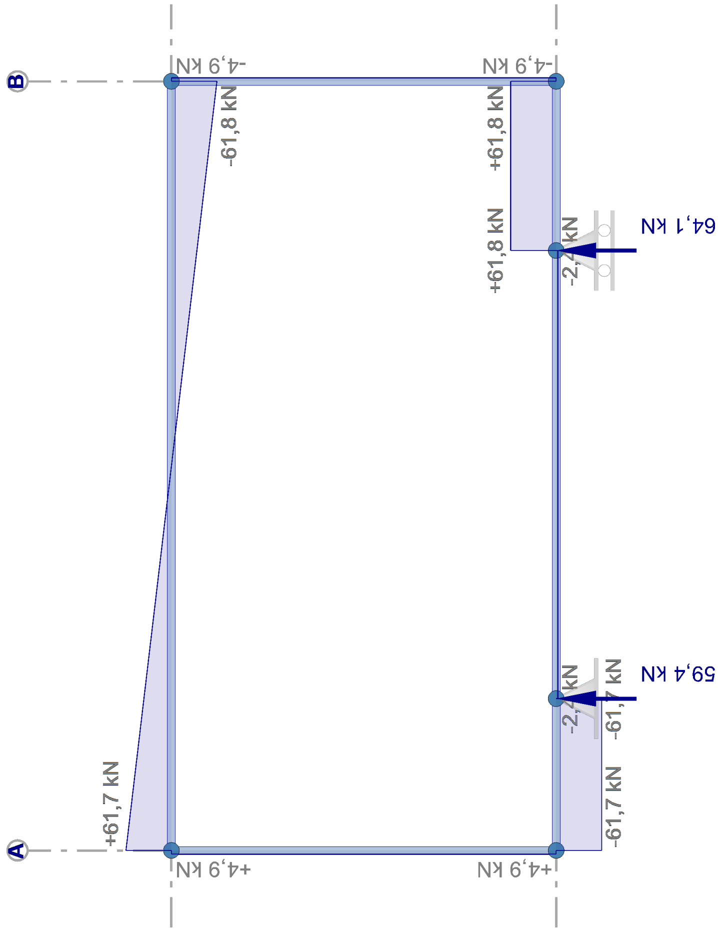
Displacements - 1 Dead load



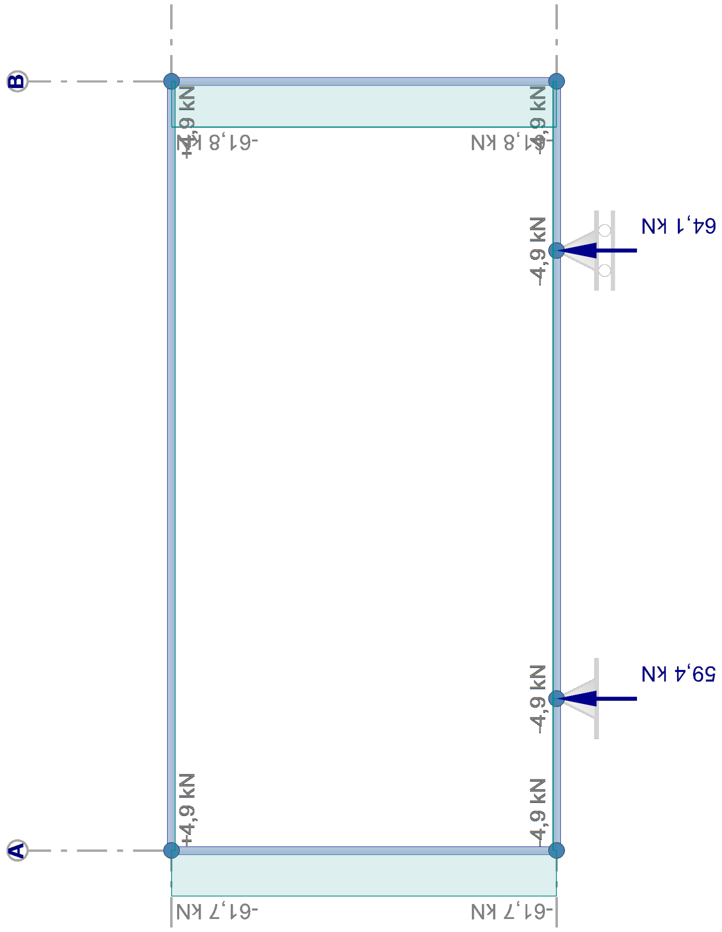
Soil pressure - 1 Dead load


B
+5,2 kNm
+17,3 kNm
-5,2 kNm
+17,3 kNm
-50,6 kNm
64,1 kN
+72 kNm
-43,7 kNm
59,4 kN
-5,1 kNm
+17,2 kNm
A
-5,1 kNm
-17,2 kNm

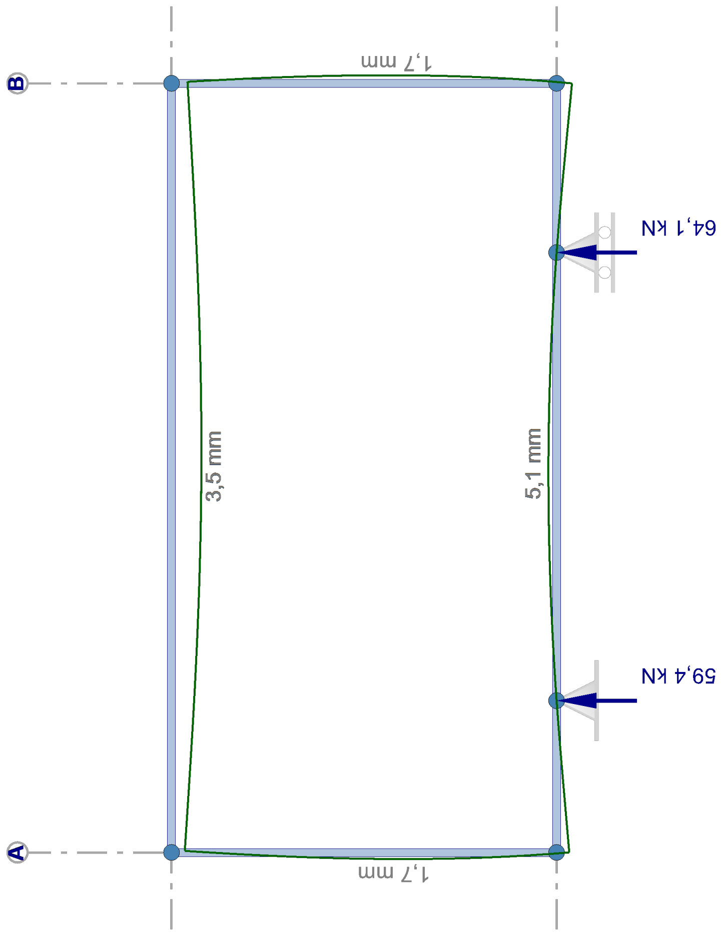
M-line - 2 Live load



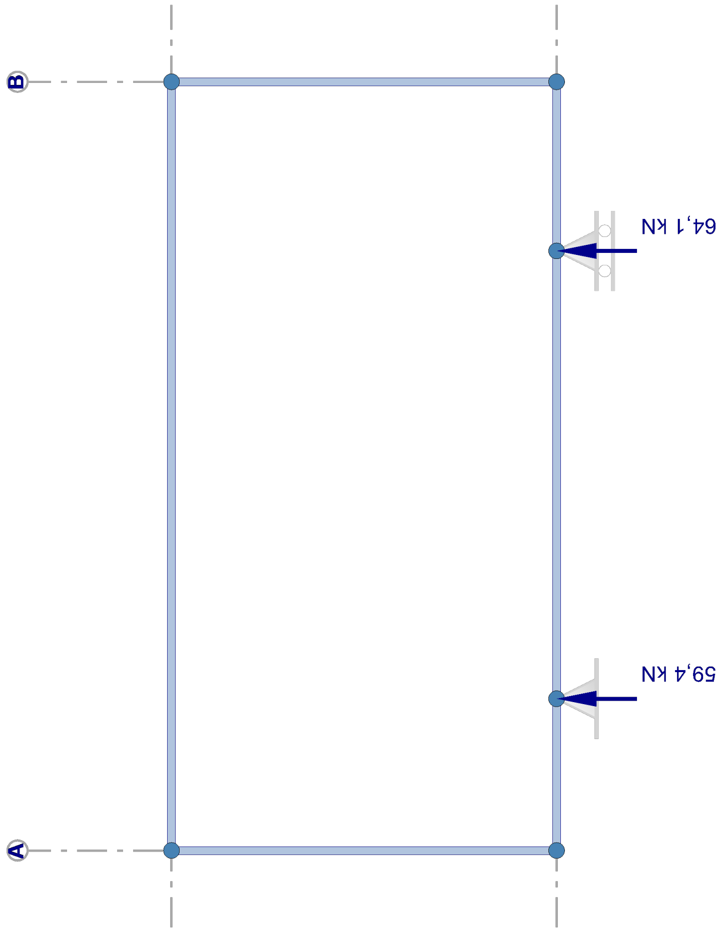
D-line - 2 Live load


B
A
+4,9 kN
-61,8 kN
-4,9 kN
-61,8 kN
64,1 kN
-4,9 kN
59,4 kN
-4,9 kN
+4,9 kN
-4,9 kN
-61,7 kN
-61,7 kN

N-line - 2 Live load



Displacements - 2 Live load


B
A
64,1 kN
59,4 kN

2.2.1 Node displacemer

Node Number	Load case
1	1 2
2	1 2
3	1 2
4	1 2
5	1 2
6	1 2
	Min
4	1
6	1
6	1
2	2
4	1
1	1

2.2.2 Envelope node di

Node Number	Load case
1	1 2
2	1 2
3	1 2
4	1 2
5	1 2
6	1 2
	Min
4	1
6	1
6	1
2	2
4	1
1	1

Soil pressure - 2 Live load

nts

	dx [mm]	dz [mm]	dr [mrad]
	0,0	**-6,5**	**6,8**
	0,0	**-2,9**	**3,0**
	0,0	**0,0**	**5,6**
	0,0	**0,0**	**2,5**
	0,0	**0,0**	**-5,9**
	0,0	**0,0**	**-2,6**
	0,0	**-8,0**	**-7,5**
	0,0	**-3,5**	**-3,3**
	0,8	**-6,8**	**-5,2**
	0,4	**-3,0**	**-2,3**
	0,8	**-8,3**	**4,6**
	0,4	**-3,7**	**2,1**
imum / maximum values			
	0,0		
	0,8		
		-8,3	
		0,0	
			-7,5
			6,8

splacements

	dx [mm]	dz [mm]	dr [mrad]
	0,0	**-6,5**	**6,8**
	0,0	**-2,9**	**3,0**
	0,0	**0,0**	**5,6**
	0,0	**0,0**	**2,5**
	0,0	**0,0**	**-5,9**
	0,0	**0,0**	**-2,6**
	0,0	**-8,0**	**-7,5**
	0,0	**-3,5**	**-3,3**
	0,8	**-6,8**	**-5,2**
	0,4	**-3,0**	**-2,3**
	0,8	**-8,3**	**4,6**
	0,4	**-3,7**	**2,1**
imum / maximum values			
	0,0		
	0,8		
		-8,3	
		0,0	
			-7,5
			6,8

2.2.3 Reaction forces

Node Number	Load case	Fx [kN]	Fz [kN]	My [kNm]
2	1		**134,902**	
	2		**59,355**	
3	1		**145,791**	
	2		**64,145**	
Minimum / maximum values				
2	2		**59,355**	
3	1		**145,791**	

2.2.4 Envelope reaction forces

Node Number	Load case	Fx [kN]	Fz [kN]	My [kNm]
2	1		**134,902**	
	2		**59,355**	
3	1		**145,791**	
	2		**64,145**	
Minimum / maximum values				
2	2		**59,355**	
3	1		**145,791**	

2.2.5 Beam forces

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	1		**10,909**	-138,806	-38,695
		2		-10,909	**-139,389**	**-98,593**
		2		**-10,909**	**-4,487**	**-98,594**
		3		**-10,909**	**-6,207**	**-114,170**
		3		-10,909	**139,584**	**-114,168**
		4		-10,909	-138,934	**39,016**
	2	1		4,852	-61,722	**-17,209**
		2		-4,852	-61,722	-43,710
		2		-4,852	-2,367	-43,711
		3		-4,852	-2,367	-50,606
		3		-4,852	61,778	-50,606
		4		-4,852	-61,778	17,350
2	1	1		**138,806**	**10,909**	**38,695**
		5		**-138,806**	**-10,909**	**-11,422**
	2	1		**61,722**	**4,852**	**17,209**
		5		**-61,722**	**-4,852**	**-5,079**
3	1	4		**138,934**	**-10,909**	**-39,016**
		6		**-138,934**	**10,909**	**11,742**
	2	4		**61,778**	**-4,852**	**-17,350**
		6		**-61,778**	**4,852**	**5,220**
4	1	5		**-10,909**	**138,806**	**11,422**
			2499	10,909	0,000	**162,006**
		6		**10,909**	**138,934**	**-11,742**
	2	5		**-4,852**	**61,722**	**5,079**

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
4			2499	4,852	0,000	72,038
		6		**4,852**	**61,778**	**-5,220**

2.2.6 Envelope beam forces

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	1		**10,909**	-138,806	-38,695
	2	1		4,852	-61,722	**-17,209**
	1	2		-10,909	**-139,389**	-98,593
	1	2		**-10,909**	-4,487	-98,594
	1	3		**-10,909**	-6,207	**-114,170**
	1	3		-10,909	**139,584**	-114,168
	1	4		-10,909	-138,934	**39,016**
2	1	1		**138,806**	**10,909**	**38,695**
	2	1		**61,722**	**4,852**	**17,209**
	1	5		**-138,806**	**-10,909**	**-11,422**
	2	5		**-61,722**	**-4,852**	**-5,079**
3	1	4		**138,934**	**-10,909**	**-39,016**
	2	4		**61,778**	**-4,852**	**-17,350**
	1	6		**-138,934**	**10,909**	**11,742**
	2	6		**-61,778**	**4,852**	**5,220**
4	1	5		**-10,909**	**138,806**	**11,422**
	2	5		**-4,852**	**61,722**	**5,079**
	1		2499	10,909	0,000	**162,006**
	1	6		**10,909**	**138,934**	**-11,742**
	2	6		**4,852**	**61,778**	**-5,220**

2.2.7 Beam stresses

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm²]	sigma2 [N/mm²]
1	1	1	0	10,909	-138,806	38,695	-54,8	51,9
			279	10,909	-138,971	0,000	-1,4	-1,4
			987	10,909	-4,487	-98,594	134,5	-137,3
1	2	1	0	4,852	-61,722	17,209	-24,4	23,1
			279	4,852	-61,722	0,000	-0,6	-0,6
			987	4,852	-2,367	-43,711	59,6	-60,9
2	1	1	0	138,806	10,909	-38,695	98,6	-149,7
			2500	138,806	10,909	-11,422	11,1	-62,2
2	2	1	0	61,722	4,852	-17,209	43,9	-66,6
			2500	61,722	4,852	-5,079	4,9	-27,7
3	1	4	0	138,934	-10,909	39,016	-150,8	99,6
			2500	138,934	-10,909	11,742	-63,3	12,1
3	2	4	0	61,778	-4,852	17,350	-67,1	44,3
			2500	61,778	-4,852	5,220	-28,1	5,4
4	1	5	0	-10,909	138,806	-11,422	7,5	-6,1
			2499	-10,909	0,000	162,006	-95,8	97,3
			5000	-10,909	-138,934	-11,742	7,7	-6,3
4	2	5	0	-4,852	61,722	-5,079	3,4	-2,7

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm²]	sigma2 [N/mm²]
4	2	5	2499	-4,852	0,000	72,038	-42,6	43,3
			5000	-4,852	-61,778	-5,220	3,4	-2,8

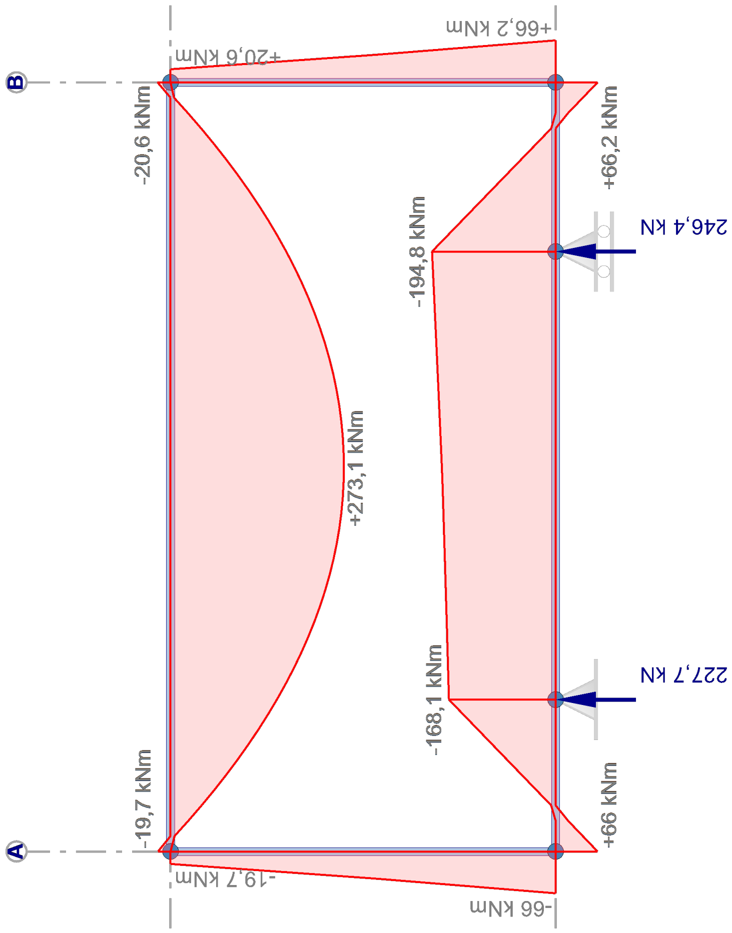
2.3 ULTIMATE LIMIT STATES (ULS)

2.3.1 Load combinations

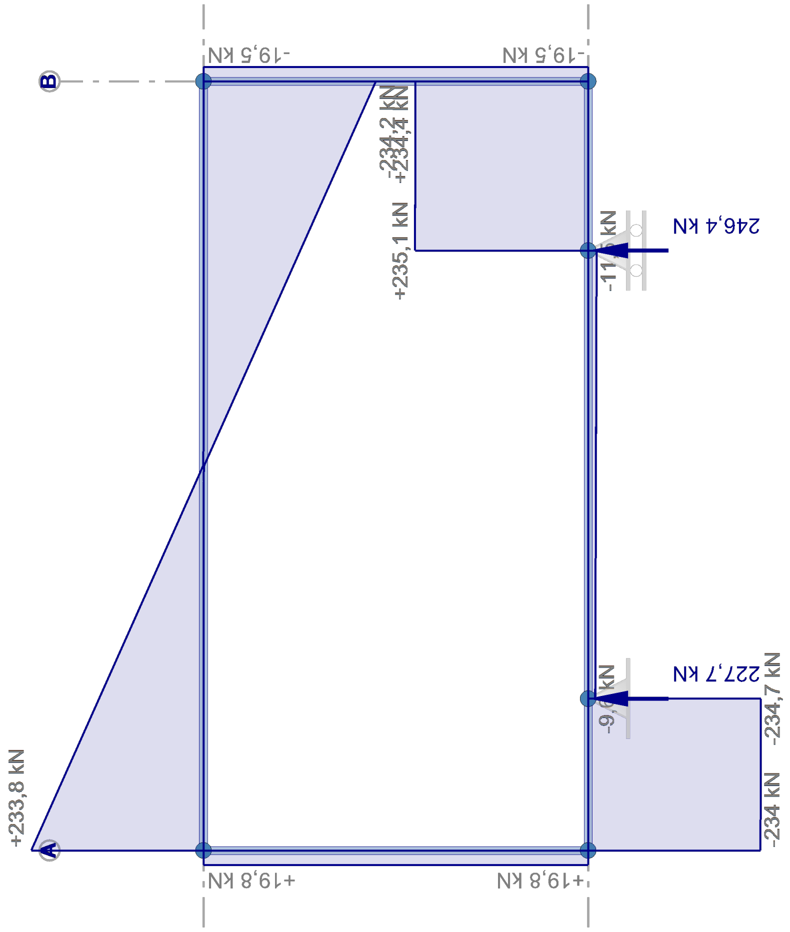
Geometric nonlinear analysis

Combination Number	Description	Type
1.1	Combination1 (6.10a) + Sway imperfection 1/200 +X	ULS
1.2	Combination1 (6.10a) + Sway imperfection 1/200 -X	ULS
2.1	Combination2 (6.10b) + Sway imperfection 1/200 +X	ULS
2.2	Combination2 (6.10b) + Sway imperfection 1/200 -X	ULS

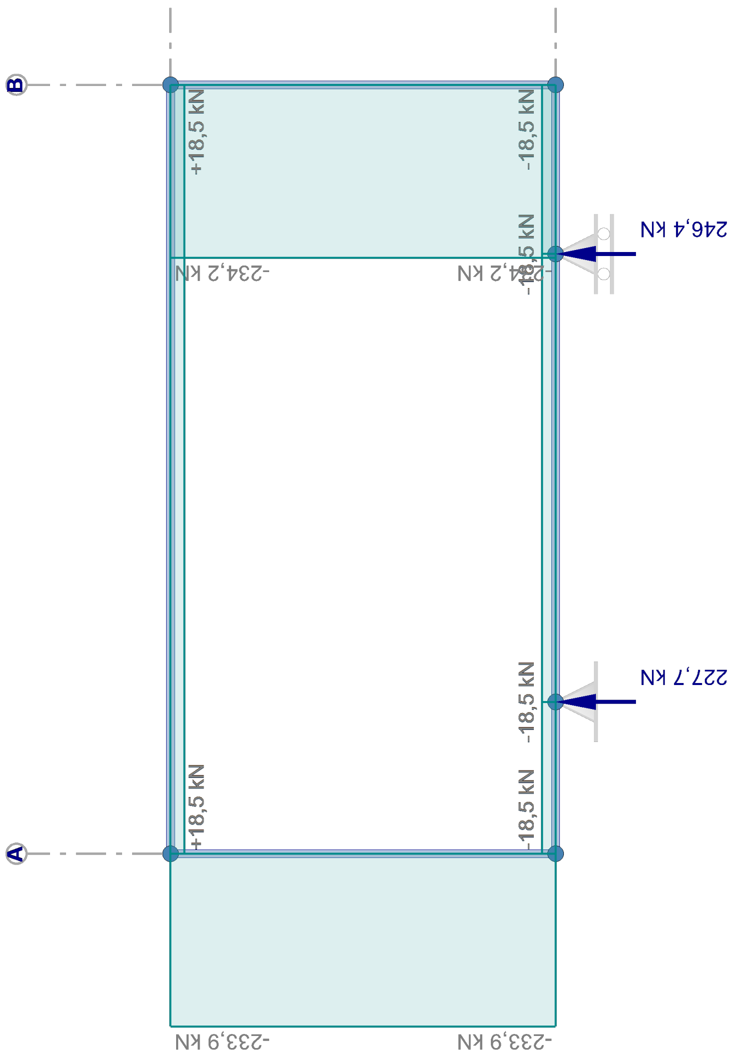
Combination Number	Load case (ψ x γ)				
	1	2			
1.1	1,00x1,22	0,40x1,35			
1.2	1,00x1,22	0,40x1,35			
2.1	1,00x1,08	1,00x1,35			
2.2	1,00x1,08	1,00x1,35			



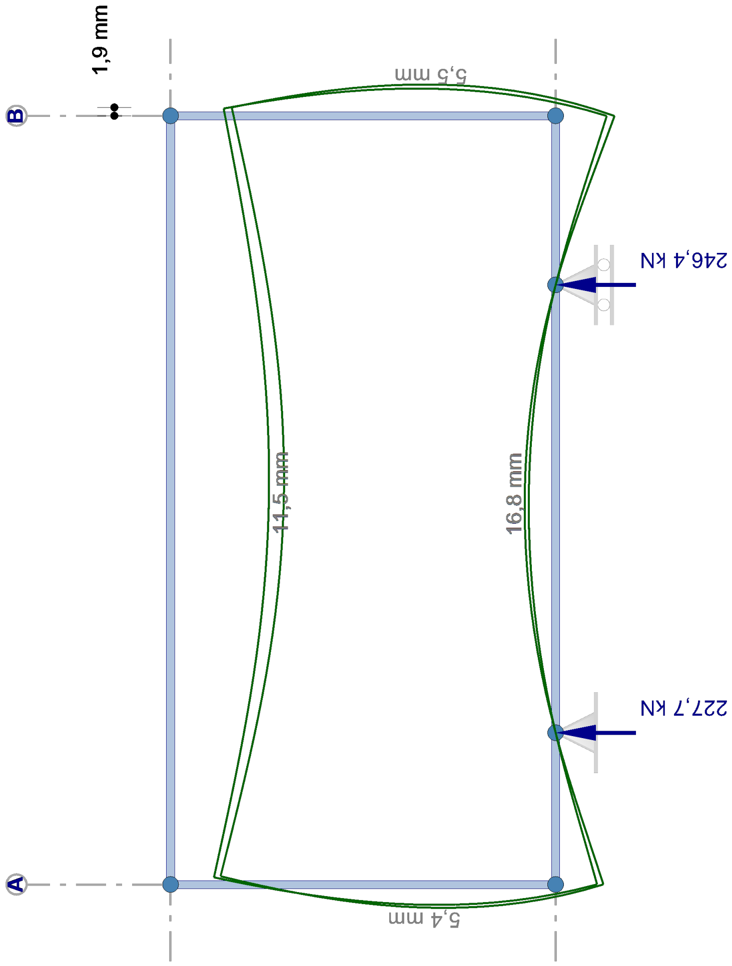
Envelop M-line


-19,5 kN
-19,5 kN
B
-234,2 kN
+234,4 kN
+235,1 kN
246,4 kN
227,7 kN
-234,7 kN
-234 kN
+233,8 kN
A
+19,8 kN
+19,8 kN

Envelop D-line



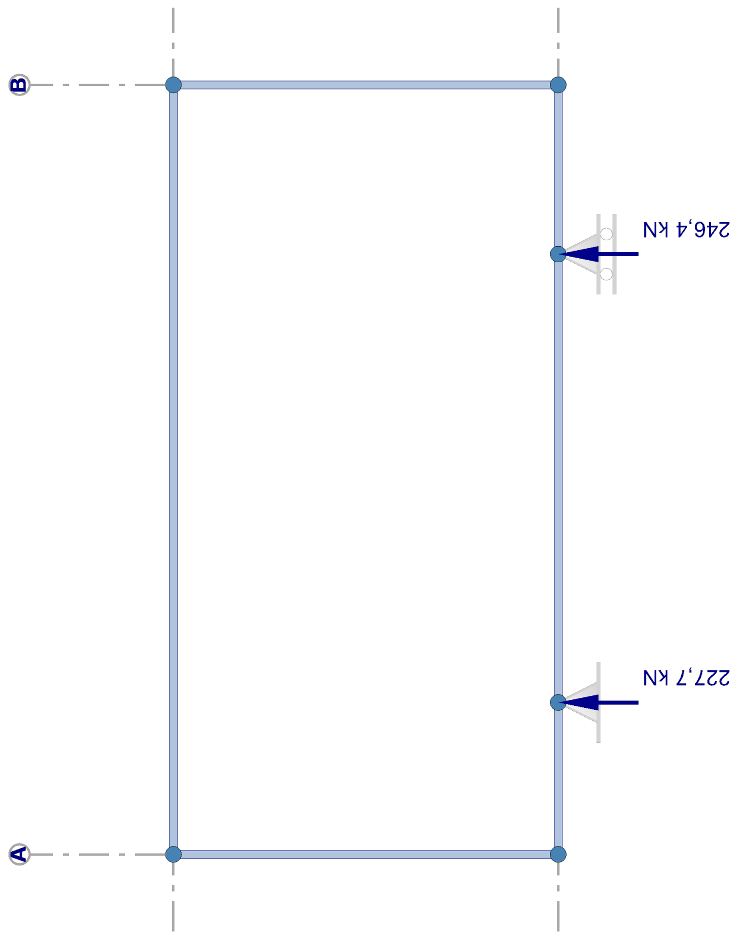
Envelop N-line


1,9 mm
5,5 mm
B
246,4 kN
11,5 mm
16,8 mm
227,7 kN
A
5,4 mm

2.3.2 Node displaceme

Node Number	Combina Numbe
1	1.1
	1.2
	2.1
	2.2
2	1.1
	1.2
	2.1
	2.2
3	1.1
	1.2
	2.1
	2.2
4	1.1
	1.2
	2.1
	2.2

Envelop displacements



Envelop soil pressure

nts

ation er	dx [mm]	dz [mm]	dr [mrad]
	0,0	**-9,5**	**9,8**
	0,0	-9,7	10,0
	0,0	-11,0	11,3
	0,0	**-11,1**	**11,5**
	0,0	**0,0**	**8,2**
	0,0	0,0	8,3
	0,0	0,0	9,5
	0,0	**0,0**	**9,6**
	0,0	0,0	-8,7
	0,0	**0,0**	**-8,7**
	0,0	**0,0**	**-10,1**
	0,0	0,0	-10,0
	0,0	-11,8	-11,1
	0,0	**-11,6**	**-10,9**
	0,0	**-13,5**	**-12,8**
	0,0	-13,4	-12,5

Node Number	Combination Number	dx [mm]	dz [mm]	dr [mrad]
5	1.1	1,6	**-10,0**	-7,7
	1.2	**0,8**	-10,1	**-7,6**
	2.1	**1,9**	-11,5	**-8,9**
	2.2	0,9	**-11,6**	-8,7
6	1.1	1,7	-12,2	**6,7**
	1.2	**0,8**	**-12,0**	6,8
	2.1	**1,9**	**-14,1**	7,8
	2.2	0,9	-13,9	**7,9**
Minimum / maximum values				
4	2.1	**0,0**		
6	2.1	**1,9**		
6	2.1		**-14,1**	
2	1.1		**0,0**	
4	2.1			**-12,8**
1	2.2			**11,5**

2.3.3 Envelope node displacements

Node Number	Combination Number	dx [mm]	dz [mm]	dr [mrad]
1	1.1	0,0	**-9,5**	**9,8**
	1.2	**0,0**	-9,7	10,0
	2.1	**0,0**	-11,0	11,3
	2.2	0,0	**-11,1**	**11,5**
2	1.1	**0,0**	**0,0**	**8,2**
	2.1	**0,0**	0,0	9,5
	2.2	0,0	**0,0**	**9,6**
3	1.2	**0,0**	**0,0**	**-8,7**
	2.1	**0,0**	**0,0**	**-10,1**
4	1.2	**0,0**	**-11,6**	**-10,9**
	2.1	**0,0**	**-13,5**	**-12,8**
5	1.1	1,6	**-10,0**	-7,7
	1.2	**0,8**	-10,1	**-7,6**
	2.1	**1,9**	-11,5	**-8,9**
	2.2	0,9	**-11,6**	-8,7
6	1.1	1,7	-12,2	**6,7**
	1.2	**0,8**	**-12,0**	6,8
	2.1	**1,9**	**-14,1**	7,8
	2.2	0,9	-13,9	**7,9**
Minimum / maximum values				
4	2.1	**0,0**		
6	2.1	**1,9**		
6	2.1		**-14,1**	
2	1.1		**0,0**	
4	2.1			**-12,8**
1	2.2			**11,5**

2.3.4 Reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
2	1.1		**194,660**	
	1.2		198,262	
	2.1		223,511	
	2.2		**227,679**	
3	1.1		214,475	
	1.2		**210,873**	
	2.1		**246,362**	
	2.2		242,194	
Minimum / maximum values				
2	1.1		**194,660**	
3	2.1		**246,362**	

2.3.5 Envelope reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
2	1.1		**194,660**	
	2.2		**227,679**	
3	1.2		**210,873**	
	2.1		**246,362**	
Minimum / maximum values				
2	1.1		**194,660**	
3	2.1		**246,362**	

2.3.6 Beam forces

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1.1	1		16,042	-202,226	-57,423
		2		-16,042	-202,936	-142,522
		2		-16,042	-8,122	-142,524
		3		-16,042	-10,220	-169,239
		3		-16,042	204,427	-169,237
		4		-16,042	-203,635	55,195
	1.2	1		16,013	-203,344	**-54,920**
		2		-16,013	-204,055	-146,129
		2		-16,013	-5,636	-146,131
		3		-16,013	-7,734	-165,604
		3		-16,013	203,308	-165,602
		4		-16,013	-202,515	57,598
	2.1	1		**18,479**	-232,736	-66,036
		2		**-18,479**	**-233,365**	**-163,982**
		2		-18,479	-9,648	-163,985
		3		-18,479	-11,506	**-194,795**
		3		-18,479	**235,084**	**-194,793**
		4		-18,479	-234,382	63,411
	2.2	1		18,445	-234,029	-63,139
		2		-18,445	**-234,659**	**-168,156**

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1		2		-18,445	-6,772	-168,159
		3		-18,445	-8,629	-190,589
		3		-18,445	233,789	-190,587
		4		-18,445	-233,088	**66,192**
2	1.1	1		**201,988**	17,184	57,423
		5		**-201,988**	-17,184	**-14,462**
	1.2	1		203,265	**15,058**	**54,920**
		5		-203,265	**-15,058**	-17,276
	2.1	1		232,435	**19,817**	**66,036**
		5		-232,435	**-19,817**	-16,493
	2.2	1		**233,911**	17,357	63,139
		5		**-233,911**	-17,357	**-19,747**
3	1.1	4		203,541	**-14,886**	**-55,195**
		6		-203,541	**14,886**	17,979
	1.2	4		**202,264**	-16,957	-57,598
		6		**-202,264**	16,957	**15,204**
	2.1	4		**234,244**	-17,124	-63,411
		6		**-234,244**	17,124	**20,602**
	2.2	4		232,768	**-19,519**	**-66,192**
		6		-232,768	**19,519**	17,393
4	1.1	5		-16,042	**202,063**	**14,462**
			2491	16,042	0,000	237,241
		6		16,042	203,470	-17,979
	1.2	5		**-16,013**	203,181	17,276
			2505	16,013	0,000	237,220
		6		**16,013**	**202,353**	**-15,204**
	2.1	5		**-18,479**	232,521	16,493
			2491	18,479	0,000	**273,135**
		6		**18,479**	**234,164**	**-20,602**
	2.2	5		-18,445	**233,813**	**19,747**
			2505	18,445	0,000	273,110
		6		18,445	232,872	-17,393

2.3.7 Envelope beam forces

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1.2	1		16,013	-203,344	**-54,920**
	2.1	1		**18,479**	-232,736	-66,036
	2.1	2		**-18,479**	-233,365	-163,982
	2.1	3		-18,479	-11,506	**-194,795**
	2.1	3		-18,479	**235,084**	-194,793
	2.2	2		-18,445	**-234,659**	-168,156
	2.2	4		-18,445	-233,088	**66,192**
2	1.1	1		**201,988**	17,184	57,423
	1.2	1		203,265	**15,058**	**54,920**
	2.1	1		232,435	**19,817**	**66,036**
	2.2	1		**233,911**	17,357	63,139
	1.1	5		**-201,988**	-17,184	**-14,462**
	1.2	5		-203,265	**-15,058**	-17,276
	2.1	5		-232,435	**-19,817**	-16,493

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
2	2.2	5		**-233,911**	-17,357	**-19,747**
3	1.1	4		203,541	**-14,886**	**-55,195**
	1.2	4		**202,264**	-16,957	-57,598
	2.1	4		**234,244**	-17,124	-63,411
	2.2	4		232,768	**-19,519**	**-66,192**
	1.1	6		-203,541	**14,886**	17,979
	1.2	6		**-202,264**	16,957	**15,204**
	2.1	6		**-234,244**	17,124	**20,602**
	2.2	6		-232,768	**19,519**	17,393
4	1.1	5		-16,042	**202,063**	**14,462**
	1.2	5		**-16,013**	203,181	17,276
	2.1	5		**-18,479**	232,521	16,493
	2.2	5		-18,445	**233,813**	**19,747**
	2.1		2491	18,479	0,000	**273,135**
	1.2	6		**16,013**	**202,353**	**-15,204**
	2.1	6		**18,479**	**234,164**	**-20,602**

2.3.8 Beam stresses

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm²]	sigma2 [N/mm²]
1	1.1	1	0	16,042	-202,226	57,423	-81,3	77,1
			284	16,042	-202,430	0,000	-2,1	-2,1
			987	16,042	-8,122	-142,524	194,4	-198,6
1	1.2	1	0	16,013	-203,344	54,920	-77,8	73,6
			270	16,013	-203,538	0,000	-2,1	-2,1
			987	16,013	-5,636	-146,131	199,4	-203,5
1	2.1	1	0	18,479	-232,736	66,036	-93,4	88,6
			284	18,479	-232,917	0,000	-2,4	-2,4
			987	18,479	-9,648	-163,985	223,7	-228,5
1	2.2	1	0	18,445	-234,029	63,139	-89,4	84,6
			270	18,445	-234,201	0,000	-2,4	-2,4
			987	18,445	-6,772	-168,159	229,4	-234,2
2	1.1	1	0	201,988	17,184	-57,423	147,1	-221,5
			2500	201,988	17,184	-14,463	9,2	-83,6
2	1.2	1	0	203,265	15,058	-54,920	138,8	-213,7
			2500	203,265	15,058	-17,276	18,0	-92,9
2	2.1	1	0	232,435	19,817	-66,036	169,1	-254,7
			2500	232,435	19,817	-16,494	10,1	-95,8
2	2.2	1	0	233,911	17,357	-63,139	159,5	-245,7
			2500	233,911	17,357	-19,747	20,3	-106,5
3	1.1	4	0	203,541	-14,886	55,195	-214,6	139,6
			2500	203,541	-14,886	17,979	-95,2	20,2
3	1.2	4	0	202,264	-16,957	57,598	-222,1	147,6
			2500	202,264	-16,957	15,205	-86,1	11,5
3	2.1	4	0	234,244	-17,124	63,411	-246,6	160,3
			2500	234,244	-17,124	20,602	-109,3	23,0
3	2.2	4	0	232,768	-19,519	66,192	-255,3	169,5
			2500	232,768	-19,519	17,394	-98,7	12,9
4	1.1	5	0	-16,042	202,063	-14,462	9,7	-7,5
			2491	-16,042	0,000	237,241	-140,3	142,5

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm²]	sigma2 [N/mm²]
4	1.1	5	5000	-16,042	-203,470	-17,979	11,8	-9,6
4	1.2	5	0	-16,013	203,181	-17,276	11,4	-9,2
			2505	-16,013	0,000	237,220	-140,3	142,5
			5000	-16,013	-202,353	-15,204	10,1	-8,0
4	2.1	5	0	-18,479	232,521	-16,493	11,1	-8,6
			2491	-18,479	0,000	273,135	-161,5	164,0
			5000	-18,479	-234,164	-20,602	13,5	-11,0
4	2.2	5	0	-18,445	233,813	-19,747	13,0	-10,5
			2505	-18,445	0,000	273,110	-161,5	164,0
			5000	-18,445	-232,872	-17,393	11,6	-9,1

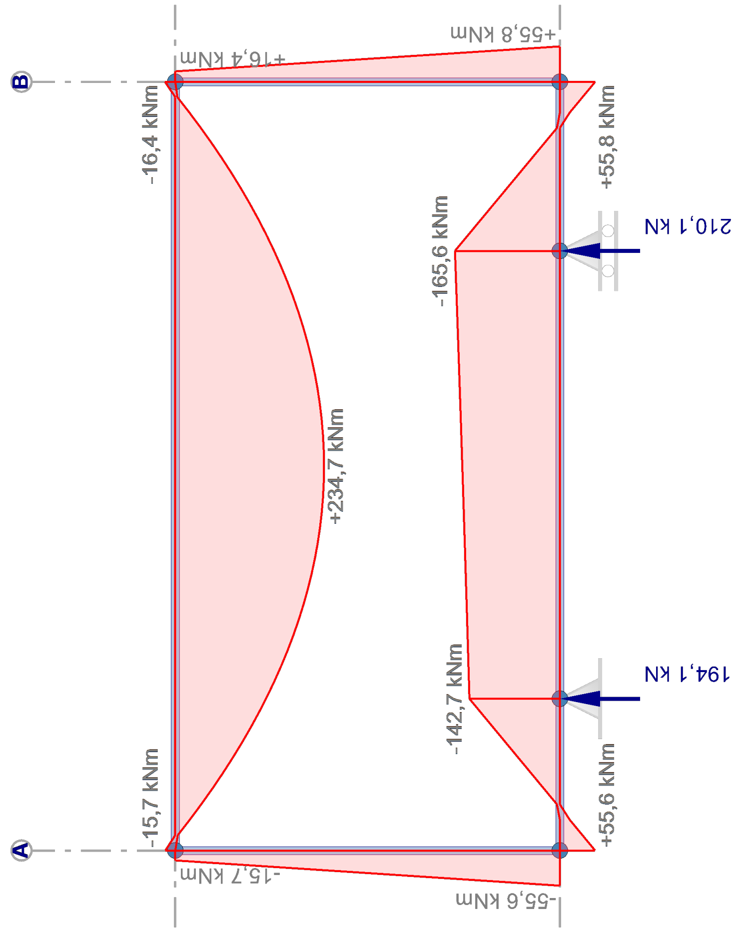
2.4 SERVICE LIMIT STATES (SLS)

2.4.1 Load combinations

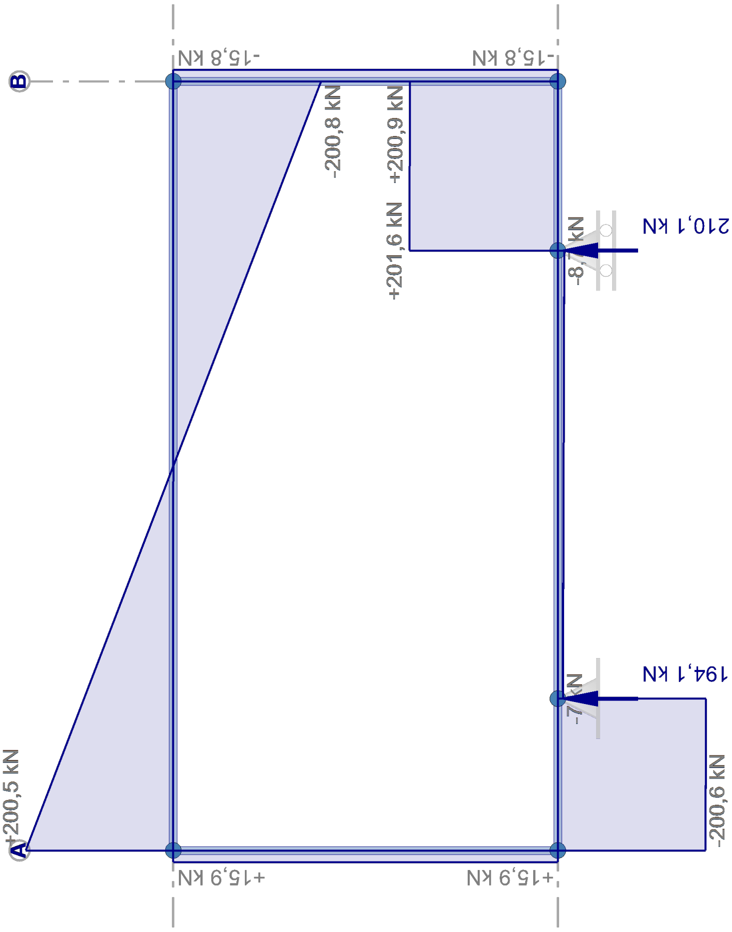
Geometric nonlinear analysis

Combination Number	Description	Type
3	SLS Permanent	SLS Permanent
4	SLS Quasi permanent	SLS Quasi permanent
5	Combination	SLS

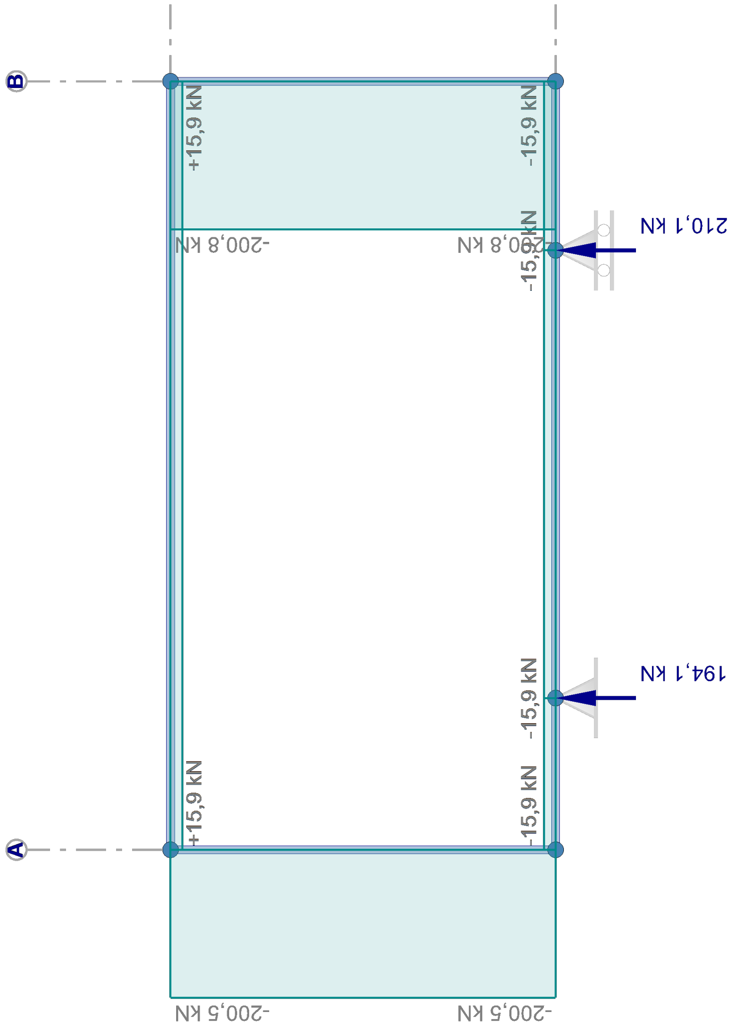
Combination Number	Load case (ψ x γ)				
	1	2			
3	1,00x1,00				
4	1,00x1,00	0,30x1,00			
5	1,00x1,00	1,00x1,00			


+55,8 kNm
+16,4 kNm
B
-16,4 kNm
+55,8 kNm
-165,6 kNm
210,1 kN
+234,7 kNm
-142,7 kNm
194,1 kN
-15,7 kNm
+55,6 kNm
A
-15,7 kNm
-55,6 kNm

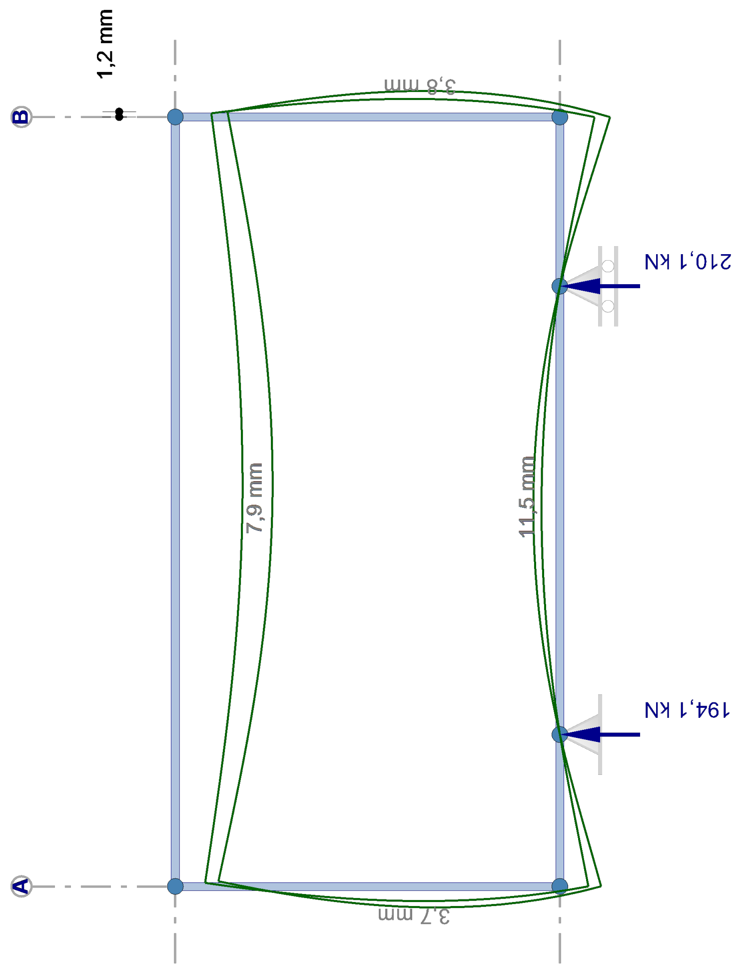
Envelop M-line



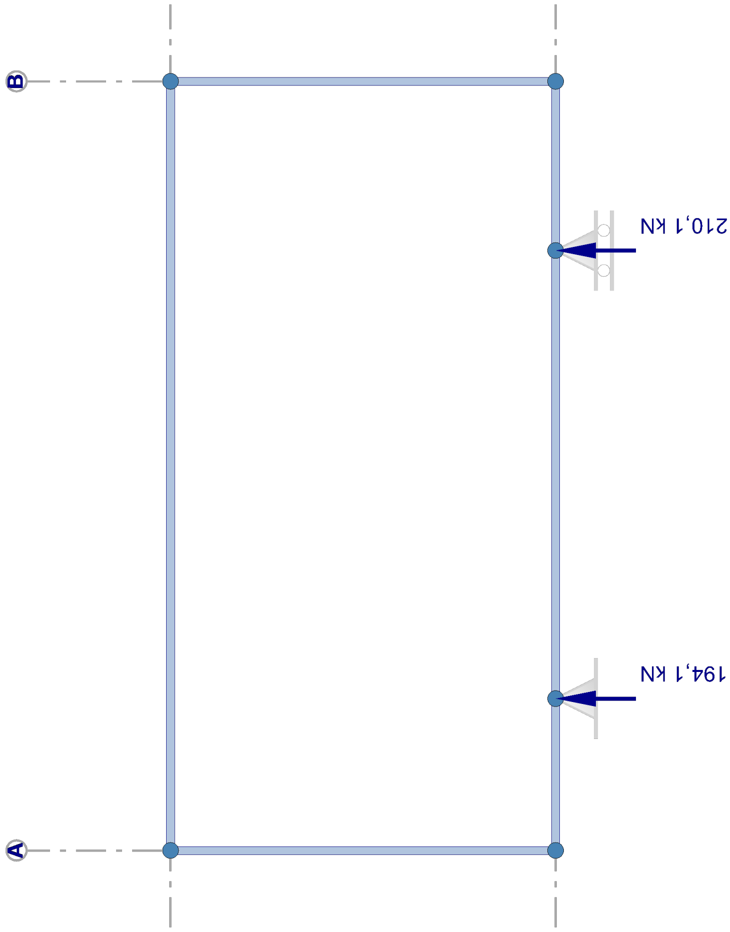
Envelop D-line


B
+15,9 kN
-15,9 kN
-200,8 kN
-200,8 kN
-15,9 kN
210,1 kN
194,1 kN
-15,9 kN
+15,9 kN
-15,9 kN
A
-200,5 kN
-200,5 kN

Envelop N-line



Envelop displacements


B
210,1 kN
194,1 kN
A

2.4.2 Node displaceme

Node Number	Combina Numbe
1	3 4 5
2	3 4 5
3	3 4 5
4	3 4 5
5	3 4 5
6	3 4 5
	Mir
4	5
6	5
6	5
2	3
4	5
1	5

2.4.3 Envelope node di

Node Number	Combina Numbe
1	3 5
2	3 4 5
3	3 5
4	3 5
5	3 5
6	3 5
	Mir
4	5
6	5
6	5
2	3

Envelop soil pressure

nts

ation er	dx [mm]	dz [mm]	dr [mrad]
	0,0	**-6,6**	**6,8**
	0,0	-7,4	7,7
	0,0	**-9,5**	**9,8**
	0,0	**0,0**	**5,7**
	0,0	0,0	6,4
	0,0	**0,0**	**8,2**
	0,0	**0,0**	**-6,0**
	0,0	0,0	-6,7
	0,0	**0,0**	**-8,6**
	0,0	**-8,0**	**-7,5**
	0,0	-9,1	-8,5
	0,0	**-11,6**	**-10,9**
	0,8	**-6,9**	**-5,2**
	0,9	-7,8	-5,9
	1,2	**-9,9**	**-7,6**
	0,8	**-8,3**	**4,7**
	0,9	-9,4	5,3
	1,2	**-12,0**	**6,7**
nimum / maximum values			
	0,0		
	1,2		
		-12,0	
		0,0	
			-10,9
			9,8

isplacements

ation er	dx [mm]	dz [mm]	dr [mrad]
	0,0	**-6,6**	**6,8**
	0,0	**-9,5**	**9,8**
	0,0	**0,0**	**5,7**
	0,0	0,0	6,4
	0,0	**0,0**	**8,2**
	0,0	**0,0**	**-6,0**
	0,0	**0,0**	**-8,6**
	0,0	**-8,0**	**-7,5**
	0,0	**-11,6**	**-10,9**
	0,8	**-6,9**	**-5,2**
	1,2	**-9,9**	**-7,6**
	0,8	**-8,3**	**4,7**
	1,2	**-12,0**	**6,7**
nimum / maximum values			
	0,0		
	1,2		
		-12,0	
		0,0	

Node Number	Combination Number	dx [mm]	dz [mm]	dr [mrad]
4	5			**-10,9**
1	5			**9,8**

2.4.4 Reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
2	3		**134,823**	
	4		152,606	
	5		**194,089**	
3	3		**145,870**	
	4		165,136	
	5		**210,103**	
Minimum / maximum values				
2	3		**134,823**	
3	5		**210,103**	

2.4.5 Envelope reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
2	3		**134,823**	
	5		**194,089**	
3	3		**145,870**	
	5		**210,103**	
Minimum / maximum values				
2	3		**134,823**	
3	5		**210,103**	

2.4.6 Beam forces

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	3	1		10,954	-138,859	**-38,541**
		2		-10,954	-139,441	-98,799
		2		-10,954	-4,546	-98,800
		3		-10,954	-6,265	-114,546
		3		-10,954	139,684	-114,545
		4		-10,954	-139,035	38,749
	4	1		12,422	-157,390	-43,659
		2		-12,422	-157,973	-111,971
		2		-12,422	-5,273	-111,972
		3		-12,422	-6,992	-129,836
		3		-12,422	158,246	-129,834
		4		-12,422	-157,597	43,877
	5	1		**15,855**	-200,637	-55,582
		2		-15,855	**-201,220**	**-142,732**
		2		-15,855	-6,978	-142,734
		3		-15,855	-8,698	**-165,565**

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1		3		**-15,855**	**201,572**	**-165,563**
		4		**-15,855**	-200,923	**55,807**
2	3	1		**138,786**	**10,999**	**38,541**
		5		**-138,786**	**-10,999**	**-11,044**
	4	1		157,296	12,480	43,659
		5		-157,296	-12,480	-12,460
	5	1		**200,485**	**15,949**	**55,582**
		5		**-200,485**	**-15,949**	**-15,710**
3	3	4		**138,955**	**-10,906**	**-38,749**
		6		**-138,955**	**10,906**	**11,483**
	4	4		157,495	-12,361	-43,877
		6		-157,495	12,361	12,976
	5	4		**200,756**	**-15,754**	**-55,807**
		6		**-200,756**	**15,754**	**16,422**
4	3	5		**-10,954**	**138,783**	**11,044**
			2498	10,954	0,000	162,324
		6		**10,954**	**138,958**	**-11,483**
	4	5		-12,422	157,292	12,460
			2498	12,422	0,000	184,027
		6		12,422	157,499	-12,976
	5	5		**-15,855**	**200,478**	**15,710**
			2498	15,855	0,000	**234,710**
		6		**15,855**	**200,763**	**-16,422**

2.4.7 Envelope beam forces

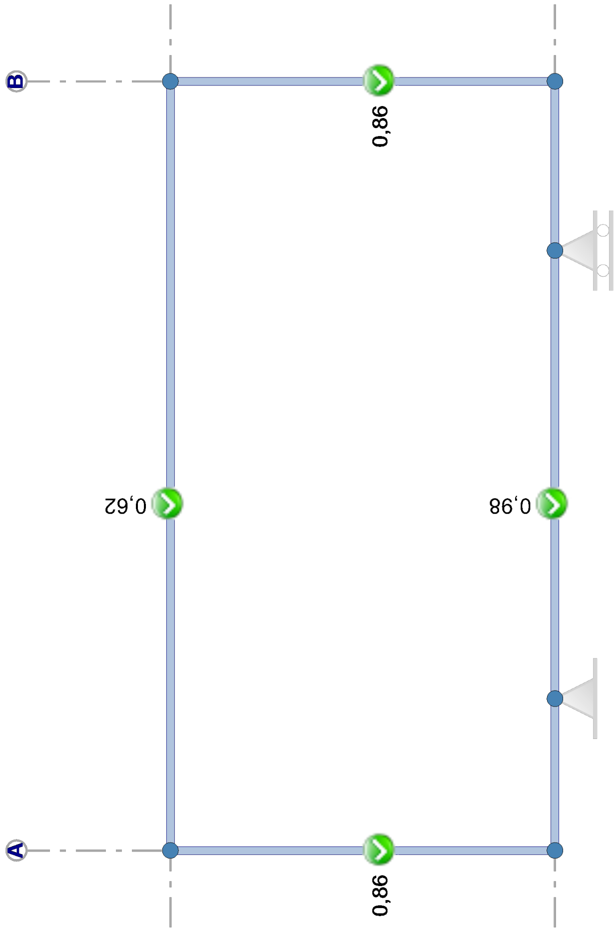
Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	3	1		10,954	-138,859	**-38,541**
	5	1		**15,855**	-200,637	-55,582
	5	2		-15,855	**-201,220**	-142,732
	5	3		-15,855	-8,698	**-165,565**
	5	3		**-15,855**	**201,572**	-165,563
	5	4		**-15,855**	-200,923	**55,807**
2	3	1		**138,786**	**10,999**	**38,541**
	5	1		**200,485**	**15,949**	**55,582**
	3	5		**-138,786**	**-10,999**	**-11,044**
	5	5		**-200,485**	**-15,949**	**-15,710**
3	3	4		**138,955**	**-10,906**	**-38,749**
	5	4		**200,756**	**-15,754**	**-55,807**
	3	6		**-138,955**	**10,906**	**11,483**
	5	6		**-200,756**	**15,754**	**16,422**
4	3	5		**-10,954**	**138,783**	**11,044**
	5	5		**-15,855**	**200,478**	**15,710**
	5		2498	15,855	0,000	**234,710**
	3	6		**10,954**	**138,958**	**-11,483**
	5	6		**15,855**	**200,763**	**-16,422**

2.4.8 Beam stresses

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm²]	sigma2 [N/mm²]
1	3	1	0	10,954	-138,859	38,541	-54,6	51,7
			277	10,954	-139,022	0,000	-1,4	-1,4
			987	10,954	-4,546	-98,800	134,8	-137,6
1	4	1	0	12,422	-157,390	43,659	-61,8	58,6
			277	12,422	-157,554	0,000	-1,6	-1,6
			987	12,422	-5,273	-111,972	152,7	-156,0
1	5	1	0	15,855	-200,637	55,582	-78,7	74,6
			277	15,855	-200,801	0,000	-2,1	-2,1
			987	15,855	-6,978	-142,734	194,7	-198,8
2	3	1	0	138,786	10,999	-38,541	98,1	-149,2
			2500	138,786	10,999	-11,044	9,9	-61,0
2	4	1	0	157,296	12,480	-43,659	111,1	-169,1
			2500	157,296	12,480	-12,460	11,0	-69,0
2	5	1	0	200,485	15,949	-55,582	141,4	-215,3
			2500	200,485	15,949	-15,710	13,5	-87,3
3	3	4	0	138,955	-10,906	38,749	-149,9	98,7
			2500	138,955	-10,906	11,483	-62,5	11,3
3	4	4	0	157,495	-12,361	43,877	-169,8	111,8
			2500	157,495	-12,361	12,976	-70,7	12,6
3	5	4	0	200,756	-15,754	55,807	-216,1	142,1
			2500	200,756	-15,754	16,422	-89,7	15,7
4	3	5	0	-10,954	138,783	-11,044	7,3	-5,8
			2498	-10,954	0,000	162,324	-96,0	97,5
			5000	-10,954	-138,958	-11,483	7,6	-6,1
4	4	5	0	-12,422	157,292	-12,460	8,3	-6,6
			2498	-12,422	0,000	184,027	-108,8	110,5
			5000	-12,422	-157,499	-12,976	8,6	-6,9
4	5	5	0	-15,855	200,478	-15,710	10,4	-8,3
			2498	-15,855	0,000	234,710	-138,8	140,9
			5000	-15,855	-200,763	-16,422	10,9	-8,7

2.5 EN1993 CHECKS

The check of the steel members in the ultimate limit state according to EN 1993-1-1 is based on a geometrical non-linear analysis (second order analysis) using the imperfections as given in 5.3.2. (a) global initial sway imperfections, see Figure 5.2)


B
A
0,86
0,62
0,98
0,86

Beam Number	Profile	Combination Number	Class	Article	U.C.
1	UNP350	2.1	1	6.2.4	0,01
		2.1	1	6.2.5	0,93
		2.1	1	6.2.6	0,35
		2.1	1	6.2.8	0,93
		2.1	1	6.3.3	0,98
		5		Deflection	0,83
		5		Deflection	0,34
2	HEB160	2.2	1	6.2.4	0,18
		2.1	1	6.2.5	0,79
		2.1	1	6.2.6	0,08
		2.1	1	6.2.8	0,79
		2.1	1	6.2.9.1	0,86
		1.1	1	6.3.2.1	0,00
		2.1	1	6.3.3	0,77
		5		Deflection	0,54
		5		Deflection	0,22
3	HEB160	2.1	1	6.2.4	0,18
		2.2	1	6.2.5	0,80
		2.2	1	6.2.6	0,08
		2.2	1	6.2.8	0,80
		2.2	1	6.2.9.1	0,86
		1.1	1	6.3.2.1	0,00
		2.2	1	6.3.3	0,78
		5		Deflection	0,54
		5		Deflection	0,22
4	HEB300	2.1	1	6.2.3	0,01
		2.1	1	6.2.5	0,62
		2.1	1	6.2.6	0,36
		2.1	1	6.2.8	0,62
		2.1	1	6.2.9.1	0,62
		1.1	1	6.3.2.1	0,00
		5		Deflection	0,57
		5		Deflection	0,23

2.6 CALCULATION OF UNITY CHECKS

2.6.1 Beam 1 - UNP350 (S 235)

Compression **art. 6.2.4**

Combination: 2.1 x = 0 mm Nx = -18,479 kN Vz = -232,736 kN My = 66,036 kNm

$$N_{c,Rd} = \frac{A\ f_y}{\gamma_{M0}} = \frac{7665{,}7 \times 235}{1{,}00} \times 10^{-3} = 1801{,}43 \text{ kN} \qquad (6.10)$$

$$\frac{N_{Ed}}{N_{c,Rd}} = \frac{18{,}5}{1801{,}4} = 0{,}01 < 1{,}0 \qquad (6.9)$$

Bending moment **art. 6.2.5**

Combination: 2.1 x = 3900 mm Nx = -18,479 kN Vz = 235,084 kN My = -194,796 kNm

$$M_{y,c,Rd} = M_{pl,y,Rd} = \frac{W_{pl,y}\ f_y}{\gamma_{M0}} = \frac{889763 \times 235}{1,00} \times 10^{-6} = 209,094 \text{ kNm} \qquad (6.13)$$

$$\frac{M_{y,Ed}}{M_{y,c,Rd}} = \frac{194,796}{209,094} = 0,93 < 1,0 \qquad (6.12)$$

Shear **art. 6.2.6**

Combination: 2.1 x = 3900 mm Nx = -18,479 kN Vz = 235,084 kN My = -194,796 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\ (f_y / \sqrt{3}\ )}{\gamma_{M0}} = \frac{4946 \times (235/\ \sqrt{3}\ )}{1,00} \times 10^{-3} = 671,1 \text{ kN} \qquad (6.18)$$

$$\frac{V_{z,Ed}}{V_{c,z,Rd}} = \frac{235,1}{671,1} = 0,35 < 1,0 \qquad (6.17)$$

Bending and shear **art. 6.2.8**

Combination: 2.1 x = 3900 mm Nx = -18,479 kN Vz = 235,084 kN My = -194,796 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\ (f_y / \sqrt{3}\ )}{\gamma_{M0}} = \frac{4946 \times (235/\ \sqrt{3}\ )}{1,00} \times 10^{-3} = 671,1 \text{ kN} \qquad (6.18)$$

$V_{z,Ed}$ = 235,084 kN < $V_{z,pl,Rd}$ / 2 = 671,060 /2 = 335,530 kN

The effect of the shear force on the moment resistance can be neglected. (2)

Uniform members in bending and axial compression (decisive) ***art. 6.3.3***

Combination: 2.1 x = 3900 mm Nx = -18,479 kN Vz = -10,613 kN My = -194,796 kNm

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \qquad \lambda_y = \frac{L_{cr,y}}{i_y}\frac{1}{\lambda_1} = \frac{5000}{128,7}\ \frac{1}{93,9} = 0,414 \qquad (6.50)$$

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \qquad \lambda_z = \frac{L_{cr,z}}{i_z}\frac{1}{\lambda_1} = \frac{5000}{27}\ \frac{1}{93,9} = 1,969 \qquad (6.50)$$

Buckling curve $_{y\text{-}y}$ c α = 0,49

$$\Phi_y = 0,5\,[1 + \alpha\ (\lambda_y - 0,2) + \lambda_y^{\,2}] = 0,5 \times [1 + 0,49 \times (0,414 - 0,2) + 0,414^{\,2}] = 0,638$$

$$\chi_y = \frac{1}{\Phi_y + \sqrt{\Phi_y^{\,2} - \lambda_y^{\,2}}} = \frac{1}{0,638 + \sqrt{0,638^{\,2} - 0,414^{\,2}}} = 0,89 \qquad (6.49)$$

Buckling curve $_{z\text{-}z}$ c α = 0,49

$\Phi_z = 0{,}5\,[1 + \alpha\,(\lambda_z - 0{,}2) + \lambda_z^2] = 0{,}5 \times [1 + 0{,}49 \times (1{,}969 - 0{,}2) + 1{,}969^2] = 2{,}872$

$$\chi_z = \frac{1}{\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}} = \frac{1}{2{,}872 + \sqrt{2{,}872^2 - 1{,}969^2}} = 0{,}201 \qquad (6.49)$$

$N_{Rk} = f_y\,A = 235 \times 7666 \times 10^{-3} = 1801{,}4$ kN

$M_{y,Rk} = f_y\,W_{pl,y} = 235 \times 889763 \times 10^{-6} = 209{,}1$ kNm

$M_{z,Rk} = f_y\,W_{pl,z} = 235 \times 139730 \times 10^{-6} = 32{,}8$ kNm

$$\frac{N_{Ed}}{\frac{\chi_y N_{Rk}}{\gamma_{M1}}} + k_{yy}\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt}\frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{18{,}479}{\frac{0{,}89 \times 1801{,}43}{1{,}00}} + 1 \times \frac{194{,}796}{1 \times \frac{209{,}094}{1{,}00}} = 0{,}94 < 1 \qquad (6.61)$$

$$\frac{N_{Ed}}{\frac{\chi_z N_{Rk}}{\gamma_{M1}}} + k_{zy}\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt}\frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{18{,}479}{\frac{0{,}201 \times 1801{,}43}{1{,}00}} + 1 \times \frac{194{,}796}{1 \times \frac{209{,}094}{1{,}00}} = 0{,}98 < 1 \qquad (6.62)$$

Deflection

Combination: 5 x = 2528 mm Nx = -15,855 kN Vz = -7,888 kN My = -154,19 kNm

Local node displacements d_{z1} = -9,5 mm d_{z2} = -11,6 mm

$w_{fin.,z} = w_z - w_{Pre\text{-}camber,z} = 16{,}6 - 0 = 16{,}6$ mm

$$\frac{|w_{fin.,z}|}{w_{fin.,z,max}} = \frac{|16{,}6|}{5000 / 250} = \frac{|16{,}6|}{20} = 0{,}83 < 1{,}0$$

$w_{add.,z} = w_z - w_{SLS\ Permanent,z} = 16{,}6 - 11{,}5 = 5{,}1$ mm

$$\frac{|w_{add.,z}|}{w_{add.,z,max}} = \frac{|5{,}1|}{5000 / 333} = \frac{|5{,}1|}{15} = 0{,}34 < 1{,}0$$

2.6.2 Beam 2 - HEB160 (S 235)

Compression **art. 6.2.4**

Combination: 2.2 x = 0 mm Nx = -233,911 kN Vz = 17,357 kN My = -63,139 kNm

$$N_{c,Rd} = \frac{A\,f_y}{\gamma_{M0}} = \frac{5427{,}5 \times 235}{1{,}00} \times 10^{-3} = 1275{,}472 \text{ kN} \qquad (6.10)$$

$$\frac{N_{Ed}}{N_{c,Rd}} = \frac{233{,}9}{1275{,}5} = 0{,}18 < 1{,}0 \qquad (6.9)$$

Bending moment **art. 6.2.5**

Combination: 2.1 x = 0 mm Nx = -232,435 kN Vz = 19,817 kN My = -66,036 kNm

$$M_{y,c,Rd} = M_{pl,y,Rd} = \frac{W_{pl,y}\ f_y}{\gamma_{M0}} = \frac{354113 \times 235}{1{,}00} \times 10^{-6} = 83{,}217 \text{ kNm} \qquad (6.13)$$

$$\frac{M_{y,Ed}}{M_{y,c,Rd}} = \frac{66{,}036}{83{,}217} = 0{,}79 < 1{,}0 \qquad (6.12)$$

Shear **art. 6.2.6**

Combination: 2.1 x = 0 mm Nx = -232,435 kN Vz = 19,817 kN My = -66,036 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\ (f_y / \sqrt{3}\ )}{\gamma_{M0}} = \frac{1762 \times (235/\ \sqrt{3}\ )}{1{,}00} \times 10^{-3} = 239{,}1 \text{ kN} \qquad (6.18)$$

$$\frac{V_{z,Ed}}{V_{c,z,Rd}} = \frac{19{,}8}{239{,}1} = 0{,}08 < 1{,}0 \qquad (6.17)$$

Bending and shear **art. 6.2.8**

Combination: 2.1 x = 0 mm Nx = -232,435 kN Vz = 19,817 kN My = -66,036 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\ (f_y / \sqrt{3}\ )}{\gamma_{M0}} = \frac{1762 \times (235/\ \sqrt{3}\ )}{1{,}00} \times 10^{-3} = 239{,}1 \text{ kN} \qquad (6.18)$$

$V_{z,Ed}$ = 19,817 kN < $V_{z,pl,Rd}$ / 2 = 239,063 /2 = 119,532 kN

The effect of the shear force on the moment resistance can be neglected. (2)

Bending and axial force (decisive) ***art. 6.2.9***

Combination: 2.1 x = 0 mm Nx = -232,435 kN Vz = 19,817 kN My = -66,036 kNm
$N_{Ed} < 0{,}25\ N_{pl,Rd}$ = 0,25 x 1275,5 = 318,9 kN (6.33)

$$N_{Ed} > \frac{0{,}5\ h_w\ t_w\ f_y}{\gamma_{M0}} = \frac{0{,}5 \times 134 \times 8 \times 235}{1{,}00} \times 10^{-3} = 126 \text{ kN} \qquad (6.34)$$

$n = N_{Ed} / N_{pl,Rd} = 0{,}18$ $a = (A - 2\ b\ t_f) / A$ = (5427,5 - 2 x 160 x 13) / 5427,5 = 0,23

$M_{N,y,Rd} = M_{pl,y,Rd}$ (1-n)/(1-0,5a) = 83,217 x (1-0,18)/(1-0,5 x 0,23) = 77,049 kNm (6.36)

$$\frac{M_{y,Ed}}{M_{N,y,Rd}} = \frac{66{,}036}{77{,}049} = 0{,}86 < 1{,}0 \qquad (6.31)$$

Lateral-torsional buckling resistance **art. 6.3.2.1**

Combination: 1.1 x = 0 mm Nx = -201,988 kN Vz = 17,184 kN My = -57,423 kNm

Number of lateral restraints: 0

d' = h-t = 160-13 = 147 mm $I_w = \dfrac{(d')^2\ b^3\ t}{24} = \dfrac{147^2 x160^3 x13{,}0}{24} = 48x10^9\ mm^6$

torsional stiffness according to Roark case 26 $I_t = 313664\ mm^4$

according to NEN-EN 1993-1-1+C2+A1/NB:2016 nl figure NB.33 and NB.34:

$L_g = 2500$ mm $L_{st} = 2500$ mm

$M_{y,1,Ed} = -14{,}463$ kNm $M_{y,2,Ed} = -57{,}423$ kNm M_{yEd} ($x=L_{st}/2$= 1250 mm) = -35,943 kNm

Calculated equivalent load q = 0 kN/m

$$B^* = \frac{8\ M}{8\ |M| + q\ L_{st}^2} = \frac{8 \times -57{,}423 \times 10^6}{8 \times |-57{,}423 \times 10^6| + 0 \times 2500^2} = -1 \qquad \text{D.4.3 (3)}$$

$$\beta = \frac{M_{y,1,Ed}}{M_{y,2,Ed}} = \frac{-14{,}463}{-57{,}423} = 0{,}252 \qquad C_1 = 1{,}803 \quad C_2 = 0$$

aangrijpingspunt belasting op z = 0 mm

$L_{kip} = L_{st} = 2500$ mm

$$S = \frac{h}{2} \times \sqrt{\frac{E \times I_z}{G \times I_t}} = \frac{160}{2} \times \sqrt{\frac{210000 \times 8892613}{80769 \times 313664}} = 687\ mm \qquad \text{(NB.159)}$$

$$C = \frac{\pi \times C_1 \times L_g}{L_{kip}} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times S^2}{L_{kip}^2} \times (C_2^2 + 1)\right.} + \frac{\pi \times C_2 \times S}{L_{kip}}\right) = \qquad \text{(NB.157)}$$

$$= \frac{\pi \times 1{,}803 \times 2500}{2500} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times 687^2}{2500^2} \times (0^2 + 1)\right.} + \frac{\pi \times 0 \times 687}{2500}\right) = 7{,}481$$

$h / t_w = 160 / 8 = 20 < 75$ $\rightarrow k_{red} = 1$ (NB.153)

$$M_{cr} = k_{red} \times \frac{C}{L_g} \times \sqrt{E \times I_z \times G \times I_t} = \qquad \text{(NB.148)}$$

$$= 1 \times \frac{7{,}481}{2500} \times \sqrt{210000 \times 8892613 \times 80769 \times 313664} \times 10^{-6} = 650{,}886\ kNm$$

$$\lambda_{Lt} = \sqrt{\frac{W_y\ f_y}{M_{cr}}} = \sqrt{\frac{354113 \times 235}{650886230}} = 0{,}358 < \lambda_{Lt,0} = 0{,}4 \qquad \rightarrow \chi_{Lt} = 1{,}00$$

$\lambda_{Lt} = 0{,}358 < \lambda_{Lt,0} = 0{,}4 \rightarrow \chi_{Lt} = 1{,}00$

Uniform members in bending and axial compression **art. 6.3.3**

Combination: 2.1 x = 0 mm Nx = -232,435 kN Vz = 19,817 kN My = -66,036 kNm

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93{,}9 \qquad \lambda_y = \frac{L_{cr,y}}{i_y} \frac{1}{\lambda_1} = \frac{2500}{67{,}8} \frac{1}{93{,}9} = 0{,}393 \qquad (6.50)$$

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93{,}9 \qquad \lambda_z = \frac{L_{cr,z}}{i_z} \frac{1}{\lambda_1} = \frac{2500}{40{,}5} \frac{1}{93{,}9} = 0{,}658 \qquad (6.50)$$

Buckling curve $_{y\text{-}y}$ b $\alpha = 0{,}34$

$$\Phi_y = 0{,}5\,[1 + \alpha\ (\lambda_y - 0{,}2) + \lambda_y^2] = 0{,}5 \times [1 + 0{,}34 \times (0{,}393 - 0{,}2) + 0{,}393^2] = 0{,}61$$

$$\chi_y = \frac{1}{\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}} = \frac{1}{0{,}61 + \sqrt{0{,}61^2 - 0{,}393^2}} = 0{,}929 \qquad (6.49)$$

Buckling curve $_{z\text{-}z}$ c $\alpha = 0{,}49$

$$\Phi_z = 0{,}5\,[1 + \alpha\ (\lambda_z - 0{,}2) + \lambda_z^2] = 0{,}5 \times [1 + 0{,}49 \times (0{,}658 - 0{,}2) + 0{,}658^2] = 0{,}828$$

$$\chi_z = \frac{1}{\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}} = \frac{1}{0{,}828 + \sqrt{0{,}828^2 - 0{,}658^2}} = 0{,}751 \qquad (6.49)$$

$N_{Rk} = f_y\ A = 235 \times 5428 \times 10^{-3} = 1275{,}5$ kN

$M_{y,Rk} = f_y\ W_{pl,y} = 235 \times 354113 \times 10^{-6} = 83{,}2$ kNm

$M_{z,Rk} = f_y\ W_{pl,z} = 235 \times 169986 \times 10^{-6} = 39{,}9$ kNm

Interaction factors according to alternative method 2 (EN 1993-1-1, Annex B)

$\varphi = M_2/M_1 = -16{,}494/-66{,}036 = 0{,}25$ $\rightarrow C_{my} = 0{,}6 + 0{,}4\ \varphi = 0{,}6 + 0{,}4 \times 0{,}25 = 0{,}7 > 0{,}4$

$$k_{yy} = C_{my}\left(1 + (\lambda_y - 0{,}2)\frac{N_{Ed}}{\chi_y N_{Rk} / \gamma_{M1}}\right) = 0{,}7 \times \left(1 + (0{,}393 - 0{,}2) \times \frac{232{,}435}{0{,}929 \times 1275{,}472/1{,}00}\right) = 0{,}726$$

$k_{zy} = 0$

$$\frac{N_{Ed}}{\frac{\chi_y N_{Rk}}{\gamma_{M1}}} + k_{yy}\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt}\frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{232{,}435}{\frac{0{,}929 \times 1275{,}472}{1{,}00}} + 0{,}726 \times \frac{66{,}036}{1 \times \frac{83{,}217}{1{,}00}} = 0{,}77 < 1 \qquad (6.61)$$

$$\frac{N_{Ed}}{\frac{\chi_z N_{Rk}}{\gamma_{M1}}} + k_{zy}\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt}\frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{232{,}435}{\frac{0{,}751 \times 1275{,}472}{1{,}00}} + 0 \times \frac{66{,}036}{1 \times \frac{83{,}217}{1{,}00}} = 0{,}24 < 1 \qquad (6.62)$$

Deflection

Combination: 5 x = 917 mm Nx = -200,485 kN Vz = 15,949 kN My = -40,96 kNm

Local node displacements $d_{z1} = 0$ mm $d_{z2} = -1{,}2$ mm

$w_{fin.,z} = w_z - w_{Pre\text{-}camber,z} = 5{,}4 - 0 = 5{,}4$ mm

$$\frac{|w_{fin.,z}|}{w_{fin.,z,max}} = \frac{|5{,}4|}{2500 / 250} = \frac{|5{,}4|}{10} = 0{,}54 < 1{,}0$$

$w_{add.,z} = w_z - w_{SLS\ Permanent,z} = 5{,}4 - 3{,}7 = 1{,}6$ mm

$$\frac{|w_{add.,z}|}{w_{add.,z,max}} = \frac{|1{,}6|}{2500 / 333} = \frac{|1{,}6|}{7{,}5} = 0{,}22 < 1{,}0$$

2.6.3 Beam 3 - HEB160 (S 235)

Compression **art. 6.2.4**

Combination: 2.1 x = 0 mm Nx = -234,244 kN Vz = -17,124 kN My = 63,411 kNm

$$N_{c,Rd} = \frac{A\, f_y}{\gamma_{M0}} = \frac{5427{,}5 \times 235}{1{,}00} \times 10^{-3} = 1275{,}472 \text{ kN} \qquad (6.10)$$

$$\frac{N_{Ed}}{N_{c,Rd}} = \frac{234{,}2}{1275{,}5} = 0{,}18 < 1{,}0 \qquad (6.9)$$

Bending moment **art. 6.2.5**

Combination: 2.2 x = 0 mm Nx = -232,768 kN Vz = -19,519 kN My = 66,192 kNm

$$M_{y,c,Rd} = M_{pl,y,Rd} = \frac{W_{pl,y}\ f_y}{\gamma_{M0}} = \frac{354113 \times 235}{1{,}00} \times 10^{-6} = 83{,}217 \text{ kNm} \qquad (6.13)$$

$$\frac{M_{y,Ed}}{M_{y,c,Rd}} = \frac{66{,}192}{83{,}217} = 0{,}80 < 1{,}0 \qquad (6.12)$$

Shear **art. 6.2.6**

Combination: 2.2 x = 0 mm Nx = -232,768 kN Vz = -19,519 kN My = 66,192 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\ (f_y / \sqrt{3}\,)}{\gamma_{M0}} = \frac{1762 \times (235/\sqrt{3}\,)}{1{,}00} \times 10^{-3} = 239{,}1 \text{ kN} \qquad (6.18)$$

$$\frac{V_{z,Ed}}{V_{c,z,Rd}} = \frac{19{,}5}{239{,}1} = 0{,}08 < 1{,}0 \qquad (6.17)$$

Bending and shear **art. 6.2.8**

Combination: 2.2 x = 0 mm Nx = -232,768 kN Vz = -19,519 kN My = 66,192 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\,(f_y/\sqrt{3})}{\gamma_{M0}} = \frac{1762 \times (235/\sqrt{3})}{1{,}00} \times 10^{-3} = 239{,}1 \text{ kN} \qquad (6.18)$$

$V_{z,Ed} = 19{,}519$ kN < $V_{z,pl,Rd}$ / 2 = 239,063 /2 = 119,532 kN

The effect of the shear force on the moment resistance can be neglected. (2)

Bending and axial force (decisive) art. 6.2.9

Combination: 2.2 x = 0 mm Nx = -232,768 kN Vz = -19,519 kN My = 66,192 kNm
$N_{Ed} < 0{,}25\,N_{pl,Rd} = 0{,}25 \times 1275{,}5 = 318{,}9$ kN (6.33)

$$N_{Ed} > \frac{0{,}5\,h_w\,t_w\,f_y}{\gamma_{M0}} = \frac{0{,}5 \times 134 \times 8 \times 235}{1{,}00} \times 10^{-3} = 126 \text{ kN} \qquad (6.34)$$

$n = N_{Ed} / N_{pl,Rd} = 0{,}18$ $a = (A - 2\,b\,t_f) / A = (5427{,}5 - 2 \times 160 \times 13) / 5427{,}5 = 0{,}23$

$M_{N,y,Rd} = M_{pl,y,Rd}\,(1\text{-}n)/(1\text{-}0{,}5a) = 83{,}217 \times (1\text{-}0{,}18)/(1\text{-}0{,}5 \times 0{,}23) = 77{,}024$ kNm (6.36)

$$\frac{M_{y,Ed}}{M_{N,y,Rd}} = \frac{66{,}192}{77{,}024} = 0{,}86 < 1{,}0 \qquad (6.31)$$

Lateral-torsional buckling resistance art. 6.3.2.1

Combination: 1.1 x = 0 mm Nx = -203,541 kN Vz = -14,886 kN My = 55,195 kNm

Number of lateral restraints: 0
d' = h-t = 160-13 = 147 mm $I_w = \frac{(d')^2\,b^3\,t}{24} = \frac{147^2 \text{x} 160^3 \text{x} 13{,}0}{24} = 48 \text{x} 10^9 \text{ mm}^6$

torsional stiffness according to Roark case 26 $I_t = 313664 \text{ mm}^4$

according to NEN-EN 1993-1-1+C2+A1/NB:2016 nl figure NB.33 and NB.34:

$L_g = 2500$ mm $L_{st} = 2500$ mm

$M_{y,1,Ed} = 17{,}98$ kNm $M_{y,2,Ed} = 55{,}195$ kNm M_{yEd} (x=L_{st}/2= 1250 mm) = 36,587 kNm

Calculated equivalent load q = 0 kN/m

$$B^* = \frac{8\,M}{8\,|M| + q\,L_{st}^2} = \frac{8 \times 55{,}195 \times 10^6}{8 \times |55{,}195 \times 10^6| + 0 \times 2500^2} = 1 \qquad \text{D.4.3 (3)}$$

$$\beta = \frac{M_{y,1,Ed}}{M_{y,2,Ed}} = \frac{17{,}98}{55{,}195} = 0{,}326 \qquad C_1 = 1{,}485 \quad C_2 = 0$$

aangrijpingspunt belasting op z = 0 mm

$L_{kip} = L_{st} = 2500$ mm

$$S = \frac{h}{2} \times \sqrt{\frac{E \times I_z}{G \times I_t}} = \frac{160}{2} \times \sqrt{\frac{210000 \times 8892613}{80769 \times 313664}} = 687 \text{ mm} \qquad (NB.159)$$

$$C = \frac{\pi \times C_1 \times L_g}{L_{kip}} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times S^2}{L_{kip}^2} \times (C_2^2 + 1) \right)} + \frac{\pi \times C_2 \times S}{L_{kip}} \right) = \quad \text{(NB.157)}$$

$$= \frac{\pi \times 1{,}485 \times 2500}{2500} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times 687^2}{2500^2} \times (0^2 + 1) \right)} + \frac{\pi \times 0 \times 687}{2500} \right) = 6{,}164$$

$h / t_w = 160 / 8 = 20 < 75 \quad \rightarrow k_{red} = 1$ (NB.153)

$$M_{cr} = k_{red} \times \frac{C}{L_g} \times \sqrt{E \times I_z \times G \times I_t} = \quad \text{(NB.148)}$$

$$= 1 \times \frac{6{,}164}{2500} \times \sqrt{210000 \times 8892613 \times 80769 \times 313664} \times 10^{-6} = 536{,}275 \text{ kNm}$$

$$\lambda_{Lt} = \sqrt{\frac{W_y f_y}{M_{cr}}} = \sqrt{\frac{354113 \times 235}{536275237}} = 0{,}394 < \lambda_{Lt,0} = 0{,}4 \quad \rightarrow \chi_{Lt} = 1{,}00$$

$\lambda_{Lt} = 0{,}394 < \lambda_{Lt,0} = 0{,}4 \rightarrow \chi_{Lt} = 1{,}00$

Uniform members in bending and axial compression **art. 6.3.3**

Combination: 2.2 x = 0 mm Nx = -232,768 kN Vz = -19,519 kN My = 66,192 kNm

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93{,}9 \qquad \lambda_y = \frac{L_{cr,y}}{i_y} \frac{1}{\lambda_1} = \frac{2500}{67{,}8} \frac{1}{93{,}9} = 0{,}393 \quad \text{(6.50)}$$

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93{,}9 \qquad \lambda_z = \frac{L_{cr,z}}{i_z} \frac{1}{\lambda_1} = \frac{2500}{40{,}5} \frac{1}{93{,}9} = 0{,}658 \quad \text{(6.50)}$$

Buckling curve $_{y\text{-}y}$ b $\alpha = 0{,}34$

$\Phi_y = 0{,}5 [1 + \alpha (\lambda_y - 0{,}2) + \lambda_y^2] = 0{,}5 \times [1 + 0{,}34 \times (0{,}393 - 0{,}2) + 0{,}393^2] = 0{,}61$

$$\chi_y = \frac{1}{\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}} = \frac{1}{0{,}61 + \sqrt{0{,}61^2 - 0{,}393^2}} = 0{,}929 \quad \text{(6.49)}$$

Buckling curve $_{z\text{-}z}$ c $\alpha = 0{,}49$

$\Phi_z = 0{,}5 [1 + \alpha (\lambda_z - 0{,}2) + \lambda_z^2] = 0{,}5 \times [1 + 0{,}49 \times (0{,}658 - 0{,}2) + 0{,}658^2] = 0{,}828$

$$\chi_z = \frac{1}{\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}} = \frac{1}{0{,}828 + \sqrt{0{,}828^2 - 0{,}658^2}} = 0{,}751 \quad \text{(6.49)}$$

$N_{Rk} = f_y A = 235 \times 5428 \times 10^{-3} = 1275{,}5$ kN

$M_{y,Rk} = f_y W_{pl,y} = 235 \times 354113 \times 10^{-6} = 83{,}2$ kNm

$M_{z,Rk} = f_y \; W_{pl,z} = 235 \times 169986 \times 10^{-6} = 39{,}9$ kNm

Interaction factors according to alternative method 2 (EN 1993-1-1, Annex B)

$\varphi = M_2/M_1 = 17{,}394/66{,}192 = 0{,}26 \quad \rightarrow C_{my} = 0{,}6 + 0{,}4\ \varphi = 0{,}6 + 0{,}4 \times 0{,}26 = 0{,}705 > 0{,}4$

$$k_{yy} = C_{my}\left(1+(\lambda_y - 0{,}2)\frac{N_{Ed}}{\chi_y N_{Rk}/\gamma_{M1}}\right) = 0{,}705 \times \left(1+(0{,}393-0{,}2) \times \frac{232{,}768}{0{,}929 \times 1275{,}472/1{,}00}\right) = 0{,}732$$

$k_{zy} = 0$

$$\frac{N_{Ed}}{\frac{\chi_y N_{Rk}}{\gamma_{M1}}} + k_{yy}\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt}\frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{232{,}768}{\frac{0{,}929 \times 1275{,}472}{1{,}00}} + 0{,}732 \times \frac{66{,}192}{1 \times \frac{83{,}217}{1{,}00}} = 0{,}78 < 1 \qquad (6.61)$$

$$\frac{N_{Ed}}{\frac{\chi_z N_{Rk}}{\gamma_{M1}}} + k_{zy}\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt}\frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{232{,}768}{\frac{0{,}751 \times 1275{,}472}{1{,}00}} + 0 \times \frac{66{,}192}{1 \times \frac{83{,}217}{1{,}00}} = 0{,}24 < 1 \qquad (6.62)$$

Deflection

Combination: 5 x = 869 mm Nx = -200,756 kN Vz = -15,754 kN My = 42,113 kNm

Local node displacements $d_{z1} = 0$ mm $d_{z2} = -1{,}2$ mm

$w_{fin.,z} = w_z - w_{Pre\text{-}camber,z} = -5{,}4 - 0 = -5{,}4$ mm

$$\frac{|w_{fin.,z}|}{w_{fin.,z,max}} = \frac{|-5{,}4|}{2500/250} = \frac{|-5{,}4|}{10} = 0{,}54 < 1{,}0$$

$w_{add.,z} = w_z - w_{SLS\ Permanent,z} = -5{,}4 + 3{,}8 = -1{,}7$ mm

$$\frac{|w_{add.,z}|}{w_{add.,z,max}} = \frac{|-1{,}7|}{2500/333} = \frac{|-1{,}7|}{7{,}5} = 0{,}22 < 1{,}0$$

2.6.4 Beam 4 - HEB300 (S 235)

Tension **art. 6.2.3**

Combination: 2.1 x = 2491 mm Nx = 18,479 kN Vz = 0 kN My = 273,135 kNm

$$N_{pl,Rd} = \frac{A f_y}{\gamma_{M0}} = \frac{14909{,}9 \times 235}{1{,}00} \times 10^{-3} = 3503{,}8 \text{ kN} \qquad (6.6)$$

$$\frac{N_{Ed}}{N_{t,Rd}} = \frac{18{,}5}{3503{,}8} = 0{,}01 < 1{,}0 \qquad (6.5)$$

Bending moment (decisive) **art. 6.2.5**

Combination: 2.1 x = 2491 mm Nx = 18,479 kN Vz = 0 kN My = 273,135 kNm

$$M_{y,c,Rd} = M_{pl,y,Rd} = \frac{W_{pl,y}\ f_y}{\gamma_{M0}} = \frac{1868933 \times 235}{1,00} \times 10^{-6} = 439,199 \text{ kNm} \qquad (6.13)$$

$$\frac{M_{y,Ed}}{M_{y,c,Rd}} = \frac{273,135}{439,199} = 0,62 < 1,0 \qquad (6.12)$$

Shear **art. 6.2.6**

Combination: 2.1 x = 5000 mm Nx = 18,479 kN Vz = -234,164 kN My = -20,602 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\ (f_y / \sqrt{3}\ )}{\gamma_{M0}} = \frac{4745 \times (235/ \sqrt{3}\ )}{1,00} \times 10^{-3} = 643,8 \text{ kN} \qquad (6.18)$$

$$\frac{V_{z,Ed}}{V_{c,z,Rd}} = \frac{234,2}{643,8} = 0,36 < 1,0 \qquad (6.17)$$

Bending and shear **art. 6.2.8**

Combination: 2.1 x = 2491 mm Nx = 18,479 kN Vz = 0 kN My = 273,135 kNm

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v\ (f_y / \sqrt{3}\ )}{\gamma_{M0}} = \frac{4745 \times (235/ \sqrt{3}\ )}{1,00} \times 10^{-3} = 643,8 \text{ kN} \qquad (6.18)$$

$V_{z,Ed}$ = 0,000 kN < $V_{z,pl,Rd}$ / 2 = 643,789 /2 = 321,894 kN

The effect of the shear force on the moment resistance can be neglected. (2)

Bending and axial force **art. 6.2.9**

Combination: 2.1 x = 2491 mm Nx = 18,479 kN Vz = 0 kN My = 273,135 kNm
$N_{Ed} < 0,25\ N_{pl,Rd}$ = 0,25 x 3503,8 = 876 kN (6.33)

$$N_{Ed} < \frac{0,5\ h_w\ t_w\ f_y}{\gamma_{M0}} = \frac{0,5 \times 262 \times 11 \times 235}{1,00} \times 10^{-3} = 338,6 \text{ kN} \qquad (6.34)$$

Allowance need not be made for the effect of the axial force on the plastic resistance moment (4)

Lateral-torsional buckling resistance **art. 6.3.2.1**

Combination: 1.1 x = 2491 mm Nx = 16,042 kN Vz = 134,475 kN My = 237,241 kNm

Number of lateral restraints: 2
Distances lateral restraints: 1667 1667 1667

d' = h-t = 300-19 = 281 mm $I_w = \frac{(d')^2\ b^3\ t}{24} = \frac{281^2 x300^3 x19,0}{24} = 1688x10^9 \text{ mm}^6$

torsional stiffness according to Roark case 26 I_t = 1857719 mm ^{4}

according to NEN-EN 1993-1-1+C2+A1/NB:2016 nl figure NB.33 and NB.34:

L_g = 5000 mm L_{st} = 1667 mm

$M_{y,1,Ed}$ = -14,46 kNm $M_{y,2,Ed}$ = 209,662 kNm M_{yEd} (x=L_{st}/2= 833 mm) = 125,762 kNm

Calculated equivalent load q = 81,105 kN/m

$$B^* = \frac{8\,M}{8\,|M| + q\,L_{st}^2} = \frac{8 \times 209{,}662 \times 10^6}{8 \times |209{,}662 \times 10^6| + 81{,}105 \times 1667^2} = 0{,}882 \qquad \text{D.4.3 (3)}$$

$$\beta = \frac{M_{y,1,Ed}}{M_{y,2,Ed}} = \frac{-14{,}46}{209{,}662} = -0{,}069 \qquad C_1 = 1{,}582 \quad C_2 = -0{,}082$$

aangrijpingspunt belasting op z = 150 mm

L_{kip} =(1,4 - (0,8 x β)) x L_{st} = (1,4 - (0,8 x -0,069)) x 1667 = 2425 mm $\rightarrow L_{kip}$ = 2333 mm

$$S = \frac{h}{2} \times \sqrt{\frac{E \times I_z}{G \times I_t}} = \frac{300}{2} \times \sqrt{\frac{210000 \times 85628952}{80769 \times 1857719}} = 1642 \text{ mm} \qquad \text{(NB.159)}$$

$$C = \frac{\pi \times C_1 \times L_g}{L_{kip}} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times S^2}{L_{kip}^2} \times (C_2^2 + 1) \right)} + \frac{\pi \times C_2 \times S}{L_{kip}} \right) = \qquad \text{(NB.157)}$$

$$= \frac{\pi \times 1{,}582 \times 5000}{2333} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times 1642^2}{2333^2} \times (-0{,}082^2 + 1) \right)} + \frac{\pi \times -0{,}082 \times 1642}{2333} \right) = 23{,}997$$

h / t_w = 300 / 11 = 27,3 < 75 $\rightarrow k_{red}$ = 1 (NB.153)

$$M_{cr} = k_{red} \times \frac{C}{L_g} \times \sqrt{E \times I_z \times G \times I_t} = \qquad \text{(NB.148)}$$

$$= 1 \times \frac{23{,}997}{5000} \times \sqrt{210000 \times 85628952 \times 80769 \times 1857719} \times 10^{-6} = 7883{,}581 \text{ kNm}$$

$$\lambda_{Lt} = \sqrt{\frac{W_y\, f_y}{M_{cr}}} = \sqrt{\frac{1868933 \times 235}{7883580963}} = 0{,}236 < \lambda_{Lt,0} = 0{,}4 \qquad \rightarrow \chi_{Lt} = 1{,}00$$

λ_{Lt} = 0,236 < $\lambda_{Lt,0}$ = 0,4 $\rightarrow \chi_{Lt}$ = 1,00

Deflection

Combination: 5 x = 2528 mm Nx = 15,855 kN Vz = -2,36 kN My = 234,675 kNm

Local node displacements d_{z1} = -9,9 mm d_{z2} = -12 mm

$w_{fin.,z} = w_z - w_{Pre\text{-}camber,z}$ = -11,4 - 0 = -11,4 mm

$$\frac{|w_{fin.,z}|}{w_{fin.,z,max}} = \frac{|-11{,}4|}{5000 / 250} = \frac{|-11{,}4|}{20} = 0{,}57 < 1{,}0$$

$w_{add.,z} = w_{z} - w_{SLS\ Permanent,z} = -11,4 + 7,9 = -3,5$ mm

$$\frac{|w_{add.,z}|}{w_{add.,z,max}} = \frac{|-3,5|}{5000 / 333} = \frac{|-3,5|}{15} = 0,23 < 1,0$$